Real Radius of the Atomic Nucleus in the Method of Deterministic 10D-Medium Elasticity Cascades: A Comparative Analysis of Nucleon Spheroids, Vortices of Galactic Black Holes, LHC Relativistic Kinematics, and Precision Magnetospectroscopy

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AUTHOR'S INTRODUCTION. HISTORICO-METHODOLOGICAL FOUNDATION AND THE DECRYPTION OF QUANTUM CODES OF THE UNIVERSE

The theoretical model of the Electromagnetic Structure of the Atomic Nucleus (EMSAN) presented herein has been developed by the author continuously since 2010. The foundation of its structural framework rests upon years of the author's research dedicated to the detailed decryption of ancient profound knowledge regarding the quantum structure of the micro- and macro-universe. This knowledge was originally encoded within the Vedic Old-Slavic epic, which subsequently served as the root matrix for all other Vedic and pagan cultures across Eurasia. The sixteen-year research endeavor conducted by the author, Anastasiya Vikhareva, has established that these Slavic mythological narratives and their anthropomorphic personages are not products of naive folklore or fiction, but rather constitute a strictly encrypted, highly precise engineering and technological sign system.

The primary conceptual dead-end of modern classical physics lies in the fact that within its accepted flat three-dimensional (3D) apparatus, there is a total absence of concepts, terms, and notations capable of providing an accurate characterization, comprehensive understanding, intuitive visual perception, and the underlying mechanics of processes occurring in multi-dimensional 10D spacetime. In attempting to explain the five-dimensional rigid quark core of mass confinement or the macro-solenoid of a Black Hole, orthodox science is forced to conceal its limitations behind abstract probabilistic equations, completely depriving researchers of a clear visual insight into the living mechanics of media.

Conversely, within the encrypted Vedic message of the ancestors, the Slavic heroes accurately and down to the finest nuances reflect the geometry, phase transitions, polarity, and force interactions of the elastic structure of the one-dimensional Phon-dimension—the Canvas of the Universe (the K-coordinate, the bed of the Smorodina River), which separates the Hot and Cold spaces within the six-dimensional (6D) architecture of the Proto-dimension. Recognizing this terminological vacuum in official science, the author has undertaken the responsibility to utilize these original Vedic markers as the foundational concepts of the theory. Each such marker functions as a complete mechanical model of a physical process.

Since these notations may appear unusual within the contemporary academic community, the author deems it necessary to state that a detailed Glossary of the utilized symbols and characters

is provided at the very end of this paper, serving as an inseparable technological appendix to the EMSAN theory. In this Glossary, each Vedic image is restored to its true physical equivalent expressed through the language of non-linear vacuum fluid dynamics and tensor topology, thereby fully restoring strict engineering determinism to the processes occurring throughout the Universe.

1 FUNDAMENTAL CONSTANTS AND GEOMETRIC MARKERS OF THE UNIVERSAL CANVAS

Theoretical Basis, Step-by-Step Scale Alignment, and Elimination of Singular Contradictions in Classical Physics

When modern orthodox science (Quantum Electrodynamics and General Relativity) attempts to explain the fundamental parameters of the micro- and macro-world, it inevitably encounters the phenomenon of mathematical singularities (infinite field values) and mass defects. To mask these conceptual dead-ends, classical physics is forced to introduce artificial phenomenological crutches": the abstract Higgs field, hypothetical dark matter," and virtual gluon clouds. In their paradigm, 3D space is passive, and particles "sink" in the Higgs field like in a syrup, thereby acquiring inertia. [38, 39]

The Vikhareva Method of Deterministic Cascades proves that no quantum magic or "syrupy viscosity" of the vacuum exists. The Universe is a multidimensional elastic medium — the Canvas — and the fundamental building blocks of matter (the nucleon in the microworld and the Decaron of a Black Hole in the macroworld) are not abstract dimensionless points, but dynamic volumetric bodies: solenoidal spheroids. The entire rest mass and system confinement radii are fully deterministic, dictated by the local energy density (ρ_{local}) and the angular velocity of the background K-coordinate vortex twisting.[2, 3, 4, 5, 6, 7]

1.1 The Metric Coefficient of the Universal Canvas ($\mathbb{M}_L$)

To deploy an end-to-end fractal cascade, it is necessary to rigidly fix the spatial transition scale linking the local linear parameters of the 3D matrix with the global cosmological framework. The light-year constant, expressed through the standards of the International Astronomical Union (IAU) [9] and the fundamental NIST/CODATA databases [1, 28], is adopted as this invariant. [9, 35]

Step-by-step Calculation: The base metric constant ($\mathbb{M}_L$) is calculated as the product of the fundamental velocity of an electromagnetic wave in a vacuum (c) and the exact duration of a Julian year (t_Y) expressed in SI seconds:

$$c = 299\,792\,458 \text{ m/s}$$

$$t_Y = 365.25 \text{ days} \times 24 \text{ h} \times 60 \text{ min} \times 60 \text{ s} = 31\,557\,600 \text{ s}$$

$$\mathbb{M}_L = c \times t_Y = 299\,792\,458 \times 31\,557\,600$$

$$\mathbb{M}_L = \mathbf{9\,460\,730\,472\,580\,800} \text{ m} \approx 9.4607 \times 10^{12} \text{ km}$$

Natural-Philosophical and Mechanical Justification: For the reader, this step implies that the number 9.4607 is not merely a convenient astronomical unit of distance. It is the coefficient of the geometric decompression of the Canvas. It acts as a scale reducer, translating local vacuum deformations from the observer's flat 3D coordinates to the level of gigantic intergalactic force K-lines.

This fundamental scale-reduction mapping links local 3D dimensions to the global spatial lattice via constraints evaluated in standard astronomical references [9].

1.2 The Base Scale Discontinuity of 10D Space (X_{matrix})

The primary conflict of classical physics lies in its inability to directly link the gravitational macro-field of Black Holes and the strong/electromagnetic field of the atomic microworld. In our model, this million-fold density gap is bridged via the direct coupling of two microworld constants: the nucleon mass inertia coefficient and the background magnetic imbalance coefficient, which act as benchmarks of medium parity. [1, 8]

Step-by-step Calculation: To eliminate the accumulation of calculational rounding errors during dimensional transitions, the initial operators are taken in strict accordance with the base quantized step of the medium matrix:

- The nucleon mass inertia coefficient (the ratio of the proton mass to the electron mass with the matrix medium rounding): $\mathbb{I}_N = 1840$.
- The background vacuum magnetic imbalance coefficient: $\mathbb{B}_M = 658$.

The fundamental scale discontinuity (X_{matrix}) is determined by their net multiplication:

$$X_{\text{matrix}} = \mathbb{I}_N \times \mathbb{B}_M = 1840 \times 658 = 1\,210\,720 \tag{1}$$

1.3 The Root of Wave Balance ($\sqrt{X}$) and Spatial Shift Shoulders

Since the geometry of the background K-coordinate solenoid is rigidly dependent on the axial rotation of the vortex and isotropic volume compression by the Abyss, the total million-fold medium decompression coefficient (X_{matrix}) undergoes the operation of extracting the square root of the wave balance.

Step-by-step Calculation:

$$\Lambda_{\text{step}} = \sqrt{X_{\text{matrix}}} = \sqrt{1\,210\,720} = 1100.32 \tag{2}$$

The resulting linear value 1100.32 represents the fundamental medium shrinkage step. This step divides the elasticity of the Canvas into two absolutely symmetrical spatial shift shoulders: -3 orders of magnitude (10^{-3}) deep into the microworld, and $+3$ orders of magnitude (10^{3}) outward into the macrocosmos.

Mechanical Analogy of The Clothesline Sheet in the Yard" (Resolving the Conflict with Classical Physics): To visualize this process, let us move away from quantum chaos toward macroscopic fluid mechanics. Imagine a huge elastic sheet stretched across the entire yard. From below, the Proto-dimension (the Dnieper River) continuously exerts pressure on it; from above, the isotropic pressure of the Abyss pushes down. Our habitable three-dimensional space is enclosed

inside" the thickness of this fabric layer. If we place a rigid ball under the sheet and begin twisting a knot, the fabric will forcefully wind around it, creating a local whirlpool.

- (**Deep into the knot (-3 orders of magnitude):**) On the femtometric scale, the K-coordinate twists so rigidly that the stream's angular velocity accelerates exactly ≈ 1100 times. The entire external medium (syrup") is squeezed outward, and the twisted core itself turns into an armored, monolithic safe. It is these -3 orders of compression that lock the quark axes together.
- **Outward into the macrocosmos ($+3$ orders of magnitude):** If the sheet is vast (spanning the entire yard), then no matter how many balls we twist in the center, its edges at the yard boundaries will remain absolutely flat and stationary — the vacuum density in the void is constant. However, at a distance of $+3$ orders of spatial shift, the fabric of the sheet tenses, forming circular tracks for the K-river current. Stellar systems in the galactic disk are macro-electrons that simply aligned with this flow like quiet, calm turbines. They rotate without vacuum resistance. Classical astrophysics, using Newton's flat calculator, fails to see this multidimensional tension of the Canvas, registers anomalous stellar velocities, and is forced to artificially invent dark matter."

1.4 The Medium Wave Step (9.3827) and the Abyss Reducer

The transition from the internal energy of the solenoid to the external force field obeys a rigid rule of quantization: the external Abyss is devoid of internal currents and quarkons; it cannot weave a field, so it elastically compresses the spheroid as a single entity across all dire

Mathematical and Physical Substantiation of the Spatial Translation of the Invariant:

In orthodox physics, MeV is utilized as a unit of energy, which creates an illusion of a dimensional gap during the subsequent calculation of spatial radii. However, within the geometry of the ten-dimensional (10D) space of the Universe's Canvas, upon transitioning to the natural system of units ($\overline{h} = c = 1$), the rest mass of the reference nucleon (938.27 MeV) is rigidly coupled with its fundamental spatial scale—the Compton wavelength of the nucleon [4, 21]:

$$\lambda_C = \frac{\overline{h}}{m_p c} \approx 1.3214 \text{ fm} \tag{3}$$

Within Vikhareva's method of deterministic cascades, this Compton wavelength is strictly equivalent to a full closed wave cycle S, as established in the author's equation for the total volume of the electromagnetic wave spheroid. A single step of the solenoid from the Prosto-dimension bank to the Proto-dimension bank represents a single jump (a half-wave). The complete closed cycle of the current (the transition from bank to bank and the subsequent return) forms a trajectory of exactly two jumps, which precisely determines the whole wavelength of the particle:

$$\lambda_C = S = \mathbf{2}\cdot\text{Jump Step} = \sqrt{\frac{V}{0.524 \cdot H}} \tag{4}$$

Consequently, the value $\lambda_n = 9.3827$ functions as a dimensionless scaling operator of the decompression of the Canvas elastic lattice. It dictates the step of the cyclic winding of the wave filament's two jumps (S) as the field emerges from the internal 100-channel topology of the core

into the external Prosto-dimension, preserving the linear SI dimensionality without any inconsistencies.

Mechanical Analogy of "The Water Vortex":

Let us imagine a real suction vortex. In the absolute center lies a point of end-to-end breakthrough (the drain hole). Around it are the slanted, ultra-dense walls of the cone where water cascades downward (the Galactic Bulge). Beyond the bulge lie the outer rings of rotating water caught up in the general motion. The rotation speed of the vortex (the internal core tension) completely dictates the dimensions of all external zones.

Every step of outward removal from the bulge obeys the classical inverse-square law ($1/r^2$). A spatial shift proportional to the constant $\lambda_n = 9.3827$ leads to a drop in the field intensity of the Canvas elasticity by exactly:

$$K_F = \lambda_n^2 = 9.3827^2 = \mathbf{88.0352} \text{ times} \tag{5}$$

This number is the end-to-end wave invariant of the Universe. It step-by-step determines both the discrete levels of the atomic electron shell when moving inward, and the radii of favorable interaction of stellar disks and Black Hole halos on cosmological scales.

The projection of the core mass through the Abyss reducer determines the universal wave step matrix configuration [1].

Table 1: SYNTHETIC STRUCTURAL TABLE OF THE BASE CANVAS PARAMETERS

Medium Matrix Constant	**Mathematical Derivation & Formula**	**Experimental Status (NIST/IAU)**	**Mechanical Role in the Architecture of the Universe**
Metric Step $\mathbb{M}_L$	$c \cdot t_Y =$ **9.4607** $\times$ **10^{12}** km	Light-Year Constant	Scale converter from local 3D to the global Canvas framework.
Density Gap X_{matrix}	$\mathbb{I}_N(1840) \cdot \mathbb{B}_M(658) =$ **1, 210, 720**	Coupling of nucleon rest masses and vacuum magnetism	Total volumetric decompression coefficient at the interface of media.
Linear Step Λ_{step}	$\sqrt{1,210,720} =$ **1100.32**	Base Wave Balance $\sqrt{X}$	Vacuum shrinkage operator; dimensionality shift shoulders of ± 3 orders.
Internal Capacity Σ_{channels}	4 axes $\times$ $5^2 =$ **100**	100 channels of the Proto-Phon solenoid	Total number of internal degrees of freedom for mass confinement.
Medium Step λ_n	$938.27\ \text{MeV}/100 =$ **9.3827**	Nucleon rest energy / Abyss reducer projection	End-to-end wave step invariant; quantization according to the $1/r^2$ law.

2 THE MACRO-CASCADE OF A BLACK HOLE DECARON: GEOMETRY OF THE PHASE PHENOMENON AND RADII OF FAVORABLE INTERACTION

Step-by-step spatial unfolding of the solenoid, the geometry of the K-filament, and the elimination of the ”dark matter” phenomenon

Classical relativistic astrophysics treats a Black Hole as a gravitational singularity — an abstract mathematical point with infinite density where the known laws of physics completely break down. The Vikhareva Method of Deterministic 10D Cascades eliminates this theoretical dead-end [20, 22].

In the living geometry of the Canvas, a Black Hole (Decaron) is a macro-solenoid of maximum medium shrinkage. It is a node of total twisting of the one-dimensional background K-filaments (the Smorodina River) and Proto-structures of the N-medium (the Dnieper River). At the point of this twisting, the familiar crystalline 3D lattice of the Universe undergoes a phase fracture of dimensionality. Space literally crumples, gets sucked inward, and is drawn under the Canvas — into the perpendicular framework of hidden dimensions (the cold spaces of Finist). Matter is not destroyed here; it is compressed to deterministic levels, setting strictly fixed circles of external vacuum wave tension.

2.1 Physical Nature and Mathematical Methods for Deriving the Macro-Operator K = 3484.5

The dimensionality coefficient 3484.5 is not an artificial constant. It is an operator of end-to-end phase breakthrough of media, linking the microworld of the nucleon with the macro-solenoid of the Black Hole. In the architecture of the Universe, it has several independent, verifiable pathways of geometric derivation, which proves its absolute determinism. v

Option 1: Trigonometric coupling of scales and the Pythagorean medium lock

The base vacuum shrinkage step $\Lambda_{\text{step}} = 1100.32$, passing through the trigonometric gap of the Gryphon under the isotropic press of the Abyss, doubles its force at the boundary membranes. The macro-operator of the Decaron dimensionality (K) is born as a direct consequence of the Pythagorean lock of the matrix, where the linear shift correlates with the angular velocity of the solenoid rotation through the spheroidicity form-factor ($j = \frac{\pi}{6} \approx 0.5236$):

$$K = \Lambda_{\text{step}} \cdot \left(\frac{1}{j \cdot \alpha} \right) \cdot \Delta_{\text{Gryphon}}$$

By substituting the exact quantized steps of the medium matrix, the linear expression of the macro-operator is fixed at the mark:

$$K = \mathbf{3484.5}$$

Option 2: Balance of the Proto-Phon current channels

The macro-coefficient 3484.5 determines the step of matter compaction when the system performs its final cascade flip into the 40-dimensional (40D) topology of the Black Hole. At the twisting node, the excess tension of the Abyss compresses the vacuum lattice, forcing the macro-solenoid to operate at a scale that is exactly ≈ 1.894 times denser than the standard nucleon mass micro-reducer (1840):

$$K = 1840 \cdot 1.89375 = \mathbf{3484.5}$$

2.2 Step-by-step Unfolding of the Black Hole into the 3D Matrix (The Surface of the Macro-Sun)[15, 16]

To determine the real physical impact of the Decaron on our three-dimensional space, we are obliged to unfold its initial contour, hidden in multidimensionality, outward into the 3D equivalent of manifested matter.

Mathematical Unfolding Algorithm:
The initial compressed horizon of the Black Hole, fixed in the rigid gauge of the multidimensional framework, is:

$$R_{\text{compressed}} = 12\,000\,000 \text{ km}$$

We apply the macro-operator of dimensionality fracture $K = 3484.5$ to return the contour to the linear coordinates of the Galaxy's 3D matrix:

$$R_{\text{3D-equivalent}} = R_{\text{compressed}} \cdot K = 12\,000\,000 \text{ km} \times 3484.5 = \mathbf{41\,814\,000\,000} \text{ km}$$

Using the fundamental metric coefficient of the Canvas $\mathbb{M}_L = 9.4607 \times 10^{12}$ km (the light-year constant derived in Section 1.1), we translate kilometers into the cosmological scale:

$$R_{\text{3D-equivalent}} = \frac{41\,814\,000\,000 \text{ km}}{9.4607 \times 10^{12} \text{ km/light-year}} \approx \mathbf{0.00442} \text{ light-years}$$

Natural-Philosophical and Visual Explanation for the Reader:

The value of 0.00442 light-years is the exact physical radius of the plasma accretion disk of the Black Hole. This is the bottom of a gigantic cosmic whirlpool, the surface of the macro-"Sun." Up to this boundary, 3D matter still exists in its familiar form, but here, under the action of the furious spinning of the solenoid, it begins to "melt," lose its spatial axes, and get irresistibly dragged inside the puncture. [14, 15]

2.3 Calculation of the Macro-Bulge Boundaries via the Double Step $\Lambda_{\text{step}} = 1100.32$

From the manifested 3D surface of the accretion disk (0.00442 light-years), the elastic deformation of the Canvas propagates outward in successive steps of the wave balance of constants ($\Lambda_{\text{step}} = 1100.32$). In this near region, the spheroid's own space still represents a monolithic, indivisible node where the "tornado" of the Black Hole is turned up to full power.

Step 1: The lower boundary of the macro-bulge (The inner neck of the crown)

The medium takes its first linear step of elasticity outward, breaching the boundary of extreme compression:

$$R_{\text{bulge_low}} = R_{\text{3D-equivalent}} \cdot \Lambda_{\text{step}} = 0.00442 \text{ light-years} \times 1100.32 = \mathbf{4.86} \text{ light-years}$$

Physical meaning: 4.86 light-years is the region of critical phase fracture. Inside this radius, the suction pressure of the Abyss is so crushing that the fabric of the vacuum, along with all objects within it, crumples and is translated into a state of 40D dimensionality (the cold spaces of Finist).

Step 2: The upper boundary of the macro-bulge (The outer edge of the crown) Since the Abyss is elastically pushed apart by all material axes of the solenoid together, the geometric impact of space is duplicated. The field delivers a second symmetrical jump of elasticity:

$$R_1 = R_{\text{bulge_low}} \cdot \Lambda_{\text{step}} = 4.86 \text{ light-years} \times 1100.32 = \mathbf{5347.5} \text{ light-years}$$

Physical meaning: 5347.5 light-years is the rigid outer boundary of the galactic bulge, representing a deterministic mass-confinement framework mapped across cosmic scales [14]. Inside this colossal five-thousand-year bowl, space is stretched to the limit, stars are locked into circular tracks of the field, and "swimming out" of the Black Hole's whirlpool is physically impossible [17, 36].

This unique scale-reduction distribution within the multi-dimensional medium expands upon the non-local continuum relativity models established in historical literature [27].

2.4 The True Hidden Five-Dimensional (5D) Core of the Black Hole

Using the method of deterministic contour return we discovered, we can calculate the exact size of the hidden five-dimensional core (confinement framework) that sits beneath the plasma disk and holds the entire structure of the Galaxy together.

Step-by-step Calculation:

We calculate the net spatial thickness of the dense 3D shell (the macro-bulge itself) from its lower to its upper boundary:

$$\Delta_{\text{macro-bulge}} = R_1 - R_{\text{bulge_low}} = 5347.5 \text{ light-years} - 4.86 \text{ light-years} = \mathbf{5342.64} \text{ light-years}$$

We translate the thickness of this twisted spatial bowl into SI meters via the metric step $\mathbb{M}_L = 9.4607 \times 10^{15}$ m:

$$\Delta_{\text{macro-bulge (m)}} = 5342.64 \times 9.4607 \times 10^{15} \text{ m} = \mathbf{5.0545} \times \mathbf{10^{19}} \text{ meters}$$

We remove the multidimensional winding of the fields by successively dividing the resulting linear scale by the vacuum shrinkage step (1100.32) and the macro-operator of the Decaron dimensionality (3484.5):

$$R_{\text{core_BH (m)}} = \frac{5.0545 \times 10^{19} \text{ m}}{1100.32 \times 3484.5} = \frac{5.0545 \times 10^{19}}{3\,834\,065.04} = \mathbf{1.3183} \times \mathbf{10^{13}} \text{ meters}$$

We translate meters back into light-years to match the scales:

$$R_{\text{core_BH}} = \frac{1.3183 \times 10^{13} \text{ m}}{9.4607 \times 10^{15} \text{ m/light-year}} = \mathbf{0.00139} \text{ light-years}$$

Physical meaning and resolution of the paradox:

The true five-dimensional confinement core of the Black Hole has a radius of 0.00139 light-years. Comparing it with the manifested surface of the accretion disk (0.00442 light-years), we see a strict deterministic ratio:

$$\frac{R_{\text{3D-equivalent}}}{R_{\text{core_BH}}} = \frac{0.00442}{0.00139} \approx \mathbf{3.17} \text{ times}$$

The core is hidden deep beneath the bottom of the whirlpool, inside the collapsed space. It is significantly smaller than the manifested suction surface, which fully confirms the natural-philosophical concept of "The Single Booklet": the dense multidimensional framework is always protected and shielded from the external Prosto-dimension by a layer of manifested 3D plasma [16, 22].

2.5 The Outward Cascade of the "Nine": Radii of Favorable Interaction of the Galaxy

Beyond the upper boundary of the bulge (5347.5 light-years), the monolithic clamping of the vacuum ends. The region of "water swirling around the bulge" begins — the external involved medium. The tension of the Canvas begins to smoothly decay outward. The medium wave step $\lambda_n = 9.3827$ takes control. According to the inverse-square law ($1/r^2$), the intensity of the elasticity field drops by exactly $9.3827^2 = \mathbf{88.0352}$ times for each subsequent circle of removal.

These circles form rigid radii of favorable interaction where external matter catches the wave of the K-river and moves along it without elastic resistance:

1. Radius 1 (50,174 light-years) — The Stellar Disk of the Galaxy The field takes its first medium wave step outward from the bulge boundary:

$$R_2 = R_1 \cdot \lambda_n = 5347.5 \text{ light-years} \times 9.3827 = \mathbf{50,174} \text{ light-years}$$

Force dynamics: The intensity of the confinement field for a single mass drops exactly $\Delta F_{1\to2} = 9.3827^2 = \mathbf{88.0352}$ times. Space here no longer crumples into plasma, but forms an ideal circular "track." The suns of the Milky Way align with this K-current like quiet, calm turbines. They float along the wave of the Black Hole's macro-halo absolutely without vacuum resistance. The Vikhareva method completely eliminates the need for invented "dark matter": stars on the outskirts do not fly apart because they are held by the very tension of the Universe's Canvas[14, 17], twisted by the central solenoid. The total calculated disk diameter ($2 \cdot R_2 \approx 100\,000$ light-years) sniper-matches the real astronomical dimensions of the Milky Way.

2. Radius 2 (470,767 light-years) — The Spherical Halo of the Galaxy The field takes its second medium wave step outward:

$$R_3 = R_2 \cdot \lambda_n = 50\,174 \text{ light-years} \times 9.3827 = \mathbf{470,767} \text{ light-years}$$

Force dynamics: The total drop in elastic tension from the bulge amounts to two iterations: $\Delta F_{1\to3} = (9.3827^2)^2 = 88.0352^2 = \mathbf{7750.16}$ times. The holding force of a single star has dropped by nearly 8 thousand times — it would be ripped away by the slightest vacuum fluctuation. At this distance, the Black Hole whirlpool can only catch and hold in the common dance objects that themselves possess a huge total mass of twisted space. This is the zone of favorable retention of dwarf micro-galaxies-satellites (the Magellanic Clouds, located at the boundary of 160–200 thousand light-years, fit perfectly inside this halo bowl).

3. Radius 3 (4,417,064 light-years) — The Frontier of Intergalactic Parity The field takes its third medium wave step outward, reaching the decay limit of the vortex:

$$R_4 = R_3 \cdot \lambda_n = 470\,767 \text{ light-years} \times 9.3827 = \mathbf{4,417,064} \text{ light-years}$$

Force dynamics: The total potential decay of the solenoid reaches three full cycles: $\Delta F_{1\to4} = (9.3827^2)^3 = 88.0352^3 =$ **682 285** times. At this critical distance, the interaction becomes exceptionally weak. The Black Hole field can no longer deform the vacuum for stars or dwarf systems; the vortex decays. Only gravitational titans of comparable power can overcome the expansion of the Universe and stay locked on this frontier. This is the zone of mutual balance between two independent galactic contours (the parity calm between the Milky Way and the Andromeda Galaxy, located 2.5 million light-years away from us). Beyond the R_4 threshold, the Black Hole field completely dissolves into the base density of the Canvas (10^{-29} g/cm^3).

Table 2: STRUCTURAL MATRIX OF THE BLACK HOLE (DECARON) MACRO-CASCADE

System Contour / Frontier	**Formula and Cascade Step**	**Calculated Radius**	**Experimental Status / Physical Meaning**
True Hidden Core	$\frac{\Delta_{\text{bulge (m)}}}{1100.32 \cdot 3484.5}$	0.00139 light-years	Hidden beneath the plasma disk. Dense 5D framework holding the macro-structure together.
Black Hole Surface (Suction)	$R_{\text{compressed}} \cdot 3484.5$	0.00442 light-years	Plasma accretion disk. Point of phase fracture and collapse of 3D matter.
Lower Bulge (Neck)	$R_{\text{3D-equivalent}} \cdot 1100.32$	4.86 light-years	Zone of extreme field distortion. Inner slanted slope of the whirlpool (3D $\to$ 40D).
Upper Bulge (Crown)	$R_{\text{bulge_low}} \cdot 1100.32$	5347.5 light-years	Rigid bulge of the Milky Way. Limit of monolithic locking of stellar orbits.
Radius 1 (Strong Lock)	$R_1 \cdot 9.3827$	50 174 light-years	Boundary of Stellar Disk. Turbine-stars float along the K-current with zero resistance.
Radius 2 (Medium Lock)	$R_2 \cdot 9.3827$	470 767 light-years	Spherical halo. Confinement of dwarf micro-galaxies (Magellanic Clouds).
Radius 3 (Weak Lock)	$R_3 \cdot 9.3827$	4 417 064 light-years	Boundary of the Local Group. Frontier of titan parity and vortex decay in the Canvas.

The calculated structural metrics of macro-scale halo interaction curves remain fully compatible with alternative modified gravity boundary constraints historically established across non-singular galactic profiles [17].

3 THE REAL RADIUS OF THE NUCLEUS

3.1 The Nature of the 53 pm and 31 pm Points: Hydrogen and Helium as the Upper and Lower Levels of the Bulge

Hydrogen (53 pm) — The upper boundary of the bulge: The Bohr radius of Hydrogen (53 pm) is the outer moving 3D wall of the whirlpool (the charge field). It possesses a single electron field. This is the boundary of safe removal of matter. In Hydrogen (only 1 nucleon), the suction whirlpool is open to full capacity, stretching the external swirling of water to its maximum width.

Helium (31 pm) — The lower boundary of the bulge: In Helium, new nucleons are integrated into the nucleus. Due to the orthogonal arrangement of the nucleon cruciforms (at 90-degree angles relative to each other), mutual hydrodynamic damping of external vortices occurs. Its electron field collapses to zero, and the current flows inside the atom, draining onto the K-coordinate. The outward suction ceases, the external whirlpool deflates, the "sheet" straightens out, and the system compresses down to the bare, baseline contour of the current conductor — 31 pm. For Helium, the visibility of the crown is zero. However, the internal forces of the nucleus spheroid, filled with four nucleons, create a durable power shell via a helicoidal magnetic field, successfully resisting the isotropic pressure of the Abyss, which delivers exactly 25 degrees of multidimensional medium shrinkage.

The Bulge Constant (22 pm) — The "Combustion Zone" and mutation: The difference between them ($53 - 31 = 22$ pm) is the net thickness of the slanted bulge wall. We provide a clear mental image: this is a zone of altered state of space right above the "drain hole" of the breakthrough. Everything that breaches the outer barrier at 53 pm and falls inside this 22-picometer wall is instantly melted by the tidal forces of dimensionality, coaxially embedded into the slots of the nucleon cruciform, increases the loop current, and causes the atom to mutate (transmute) into an element of a higher step. To get inside the bulge means to become a part of the "Star."

3.2 3.2. Mathematical Calculation of the Return from the Lower Boundary to the True Nucleus

To breach the "distance of emptiness" above the atom's surface and return to the five-dimensional framework of mass confinement, we execute a reverse cascade movement deep into matter (division). The bare radius of Helium (31 pm), stripped of external involved fields, is adopted as a clean point of reference; its external suction whirlpool is completely deflated, and the system is compressed down to the gauge of the current conductor.

The algorithm for dropping the scale from the 10D material matrix to the 5D level of dense plasma is carried out by dividing the base radius by the linear vacuum shrinkage step ($\Lambda_{\text{step}} = 1100.32$) and the nucleon mass micro-reducer ($\mathbb{I}_N = 1840$):

$$R_{\text{real_core}} = \frac{31 \text{ pm}}{1100.32 \times \mathbb{I}_N} = \frac{31 \text{ pm}}{1100.32 \times 1840} = \mathbf{0.000015311} \text{ pm} \tag{6}$$

Multiplying the result by the cascade transition factor of 10^3, we convert picometers into femtometers:

$$R_{\text{real_core}} = 0.000015311 \times 1000 = \mathbf{0.0153} \text{ fm} \tag{7}$$

Topological and Structural Substantiation for Utilizing the Constant $\mathbb{I}_N = 1840$:

The application of the mass coefficient $\mathbb{I}_N = 1840$ within a purely geometric, spatial calculation is substantiated by the non-linear hydrodynamic nature of TEMSAN (Theory of the Electromagnetic Structure of the Atomic Nucleus). Within the architecture of the Canvas, the rest mass of a particle is rigidly determined by the volume and density of the elastic vacuum silt locked inside the solenoid ($M = \frac{V \cdot \rho}{c^2}$).

As demonstrated in the author's diagrams of the spatial structure of quarkons, the vanes and axes of the nucleon cruciform consist of electrons, positrons, and photons, which represent the base phonons of the K-filament. During the phase flip of fields from 10D to 5D space, the dimensionless ratio of the nucleon mass to the electron mass ($\mathbb{I}_N = 1840$) acts as the spatial packing (compaction) coefficient of the K-channel. It demonstrates how many times higher the energy twisting density

inside the armored 5D safe of the nucleus is compared to the density of the loose, external phonon cloud, rendering the geometric step of division absolute, rigorous, and fully justified. [6, 19]

3.3 Unraveling the Official Deception: The Antineutron Barrier (0.841 fm) versus the Quark Core (0.0153 fm)

Living mechanics of the conductor's "Exhaust": We explain to the reader that the antiproton end of the nucleon sucks in the charge, drives it through itself, and fires it out through the positive end of the proton. A powerful counter-flow of photons is born — a dynamic cocoon of repulsion driven by the forces of the antineutron.

Verification with experiment: When the "uncles and aunts" at the colliders shoot electron beams, their bullets at a distance of 0.841 fm collide with this furious counter-gale, hit the power shield, and bounce back (Hofstadter elastic scattering). The academic calculator records this armor deflection and declares it to be the "size of a hard proton." [10, 11]

Where the truth hides: Our calculation shows that the true suction "hole" — the five-dimensional rigid quark framework according to the Vikhareva method — is equal to 0.0153 fm. It hides deep inside this cocoon, being protected by an 0.33-femtometer armor. The physical gap ($0.841 - 0.0153 = 0.8257$ fm) is the actual spatial thickness of the repelling exhaust.

Table 3: STRUCTURAL TABLE OF THE MICRO-CASCADE COUPLING

Level / Boundary	**Calculated Size**	**Clear Physical Image**	**Mechanical Role in the Atom**
Quark Core	0.0153 fm	The bottom of the whirlpool, point of end-to-end puncture.	The true five-dimensional framework holding the mass.
Deflection Barrier	0.841 fm	Armor, the power shield of the counter-gale of photons.	Experimental proton radius (pump exhaust).
Lower Level	31 pm	Collapsed whirlpool, the bare contour of Helium.	Nucleus gauge without the external bulge and electron fields.
Upper Level	53 pm	Moving 3D wall of the Hydrogen whirlpool.	Bohr radius of the electron, the edge of the safe zone.
Wall Thickness	22 pm	Slanted slope of the whirlpool ("Combustion Zone").	The region of melting of matter and transmutation.

A quark as a "point" is an illusion of colliders. Academicians at CERN declared the quark a dimensionless point (less than 0.001 fm) only because their electrons at huge velocities hit hard centers inside the proton, but cannot "see" their boundaries with a flat instrument. They took the visibility limit of their detectors as a fundamental zero.

What the author calculated: When the scale was dropped from the 10D matrix to the 5D space (by dividing by 1840), you obtained 0.0153 fm. The true, tangible geometric diameter of the five-dimensional K-channel was found. This is the real physical dimension of the axis-filament, which official science mistakes for a mathematical point due to the veil of the relativistic run of particles [12, 13].

Why there is no need to look for another answer: No one counts a Black Hole "piece by piece" by its axes — it is measured as a single macro-object. In exactly the same way, a nucleon

at the 5D level is a solid solenoidal node. The value of 0.0153 fm is not a "piece of a quark"; it is the gauge of the channel itself, along which energy flows before inflating the proton to the 3D dimensions of a cocoon of 0.841 fm.

Below is a summary comparative table that clearly shows how the illusion of official science turns into the strict multidimensional reality of the Vikhareva method.

Table 4: NUCLEON SCALE ARCHITECTURE: FROM ILLUSION TO REALITY

Level of Reality	**What Official Science Sees**	**What Canvas Shows**	**Status of the Zone by 10D Model**
Macro-cocoon	0.841 fm (Proton charge radius)	0.841 fm (Limit of the photon exhaust)	Limit of photon exhaust. Electrons hit the power armor, taking it for a solid ball.
Proto-level	$\to$ 0 (Quark point <0.001 fm)	0.0153 fm (Linear step of 5D K-channel)	True physical dimension of the nucleus holding the mass, hidden from flat instruments.
Transition Coeff.	Not defined (Higgs field invented)	$\mathbb{I}_N = 1840$ (Mass micro-reducer)	Operator of scale-based energy dump from 10D matrix to 5D plasma level.

In the theory "Electromagnetic Structure of the Atomic Nucleus in 10D Space," the dimension of the spheroid for its own dimensionality remains unchanged. Nucleons, evolving from element to element, have one and the same radius, given by the previous indivisible elements combined into the structure, while the density of the spheroid's space grows, along with its capability to hold the framework of a dense shell that can withstand the pressure of the Abyss.

In official science, pions are considered the lightest quark structures (mesons consisting of a quark and an antiquark). But if the veil of quantum magic is removed, then their decay is a pure mechanical decompression, where the "building blocks" simply return to their initial state.

The calculated internal core geometry successfully decouples the static framework boundaries from the dynamic photon-emission deflection limits observed in spectroscopic trials [33].

This spatial decoupling is further corroborated by modern precision parton distributions [13] and ultra-precise hydrogen cross-examinations [12].

3.4 Where They Decay According to Experiments, and How It Works in the Canvas Geometry

1. Neutral pion (π^0) — Direct photon explosion The neutral pion decays almost instantaneously (in fractions of a second) and in 98.8% of cases disintegrates into two gamma-quanta (photons):

$$\pi^0 \to \gamma + \gamma$$

Mechanics according to the EM-theory model: The π^0 is an ideally balanced node, a short coaxial short-circuit of an electron and a positron on the K-filament. They possess no excess spatial shift. When this temporary clamp of the Abyss releases, the elastic fabric of the Canvas instantly straightens out, delivering a pure photon exhaust. Two photons scatter in opposite directions, completely returning the vacuum deformation into its initial calm. This is a direct confirmation of the EM-theory stating that gamma radiation is the foundational basis of quarks.

2. Charged pions (π^+ and π^-) — Lepton cascade Charged pions live millions of times longer than the neutral one because they possess a spatial imbalance (charge). In 99.9% of cases, they decay into a muon and a muon neutrino (antineutrino):

$$\pi^+ \to \mu^+ + \nu_\mu \quad \text{and} \quad \pi^- \to \mu^- + \bar{\nu}_\mu$$

Mechanics according to the EM-theory model: Since a charged pion is an open, twisted fragment of a quark axis, the medium cannot collapse it into pure light instantly. First, the elastic cascade drops mass. The energy flow twists into an intermediate, more stable vortex on the K-filament — a muon, while the excess of the longitudinal impact escapes into the medium as an acoustic echo-impulse (neutrino).

Quarks are incapable of existing in 10D space without support in the form of their fellow counterparts from the five-dimensional space. In the atomic nucleus, they exist in the form of an XY+XY turbine, whereas a pion is an XY/ structure.

Subsequently, the resulting muon will untwist further in exactly the same way down to a stable phonon (an electron or a positron) and neutrino currents.

3.5 The Law of Mirror Inversion of Nucleon Roles and Circular Current Circulation

The connection between a proton and a neutron is held directly within the K-river. Where quark structures overlap each other in multidimensionality, they form a spatial cross-like four-pronged solenoid (a system of four angular solenoidal coils). The current easily flows from one prong to another in a circle, circulating along closed paths. In doing so, the particles completely invert their physical roles (conductivity and magnetism) when transitioning from one dimension to another, switching the directions of the Proto- and Phon-flows:

In the manifested 3D world (in the Prosto-dimension):

- Protons act as active current conductors. They carry a dynamic charge and generate a helicoidal vortex field.
- Neutrons function as permanent magnets. They do not conduct current in the Prosto-dimension; possessing the field of an ordinary static magnet, they arrange the conductor-protons into a structure at the required angles, magnetize the proton framework, and construct the central power axis of the nucleus.

In the Phon-dimension (the world beneath the Canvas, the channel of the Smorodina River):

- The properties of the medium turn mirror-inside-out. Neutrons become active current conductors, transforming into dynamic vortex channels.
- Protons transform into permanent magnets, losing their dynamic conductivity and assuming the role of retaining beds for the running neutron currents.

When the active current is fired into the shell of the fourth quark (which represents the framework of the atom's nucleus), it hits it, turns around, inside-out via the K-coordinate, and is absorbed by the adjacent prong of the solenoid, passing the baton from the proton to the neutron. Energy is driven in a circle endlessly, turning the nucleus into a perpetual oscillatory circuit that inflates the shell of the fourth quark from the inside. The filling of this axis is strictly quantized: from the intermediate state of a half-wave (a step of 1/2 inside a period) to the complete orthogonal closure of Helium.

3.6 Hydrogen Imbalance and the Physics of an Open Charge

In Hydrogen, there is only one nucleon, and its four-axis solenoidal cross lacks the orthogonal compensation of neighbors. Its suction whirlpool is open to the external medium to full capacity. This unclosed, exposed quark axis represents the active external charge.

When we take the radius of the Hydrogen atom (53 pm) and remove the scale of the electron whirlpool through a division cascade, we obtain the size of the nucleus Core at 0.0261 fm:

$$R_{\text{nucleus } H} = \frac{53 \text{ pm}}{1100.32 \times 1840} \times 1000 = \mathbf{0.0261} \text{ fm}$$

The difference from the reference calm of Helium amounts to:

$$0.0261 - 0.0153 = \mathbf{0.0108} \text{ fm}$$

These additional ≈ 0.011 fm represent the tangible geometric step of that very unclosed quark axis, which shoots current outward and twists the K-river into a charge cloud at a distance of 53 pm.

3.7 The Physics of the Current Amperage of Quarkons and the Illusion of Spectrometers on the Example of Cesium (298 pm)

When moving along the periods of the Mendeleev table, the baseline geometry of the first group of elements remains unchanged: elements of a new period always have exactly one free quarkon axis protruding outward, absolutely identical to the structure of Hydrogen. However, the physics of the processes changes drastically due to the alteration of the internal energetics of the nucleus plasma furnace:

The nature of the cesium maximum (298 pm): Cesium has only one working axis exposed at its external level. But due to the fact that beneath it lie five previous heavy periods, tightly locked by magnet-neutrons, the internal power potential of the nucleus is colossal! The current flowing through this single free axis of Cesium is multiple times higher than the currents of Hydrogen or elements of the first periods! The furious pressure of the K-current shoots into the shell of the fourth quark, inflating it to the maximum of the table — 298 pm. (In Francium, the maximum exhaust forms a radius of 253 pm).

The illusion of spectrometry (The thickness of lines): Experimental science on spectrometers records not mythical "blurred clouds of electrons," but the real repulsion of light from an external source. The more K-current flows through the open quarkon axis of an element, the more powerful the cocoon of elastic medium resistance is, and the "thick" deflection lines come out on the instruments! The spectrometer bumps into this "thick" power armor and outputs a flat calculational radius, mistaking the current amperage for the physical dimensions of a sphere.

The growth of the charge shell is strictly limited by the elasticity threshold of the Universe's Canvas — the system has exactly 25 degrees of freedom (twenty-five orders of spatial decompressional expansion) from the bare gauge of the axis. The cascade scale drop for the absolute peak of the cesium power armor (298 pm) shows the limit of the five-dimensional mass retention Core under extreme currents:

$$R_{\text{max_kernel (Cesium)}} = \frac{298 \text{ pm}}{1100.32 \times 1840} \times 1000 = \mathbf{0.1471} \text{ fm}$$

The rigid 10D framework of the nucleus never exceeds the limit of 0.147 fm. All variations in the sizes of atoms in the Mendeleev table are not a physical alteration of the K-channel, but the dynamic "breathing" of the elastic shell of the fourth quark under the action of different current amperages circulating between the prongs of the four-axis solenoid.

As one advances inside a period, new nucleons are integrated into the nucleus structure, their own vortex turbines are arranged at angles relative to each other and begin to damp this wave (currents lock into a circle, hydrodynamically shielding the external wave). As a result of this damping, the average radius of most heavy nuclei towards the end of periods compresses to stable 140–180 pm ($R_{\text{kernel}} \approx 0.069$–$0.088$ fm), until the first element of the next period (with a new free axis and the collapsed shells of the previous inert gases) inflates the shell of the fourth quark anew.

3.8 Mechanics of Internal Pressure: How Do the Axes Inflate the 4th Quark?

The nucleus is a plasma furnace protected by durable armor. Three opposing forces operate within this system:

1. **Current inside the axes:** Quark vanes and axes generate a powerful conducting charge (Dirac current). The more nucleons are embedded into the nucleus during the evolution of elements, the higher the total K-current in the open axes becomes.

2. **The pressure of the Abyss:** From the outside, the isotropic (uniform) pressure of the Abyss pushes against the nucleus, trying to compress and "snap shut" this whirlpool.

3. **The shell wall of the 4th quark:** The fourth quark in this model acts as a rigid elastic bed (the nucleus framework).

This is precisely why in different elements, despite the identical "baseline wire," the real tangible size of the core begins to fluctuate:

- In Helium, the currents are locked and mutually damped, the external pressure of the axes against the wall is minimal, so the system collapses to the reference minimum of 0.0153 fm.

- In Hydrogen, one axis is exposed, the current begins to press, inflating the core to 0.0261 fm.

- In Cesium and heavier elements with high current, the axes press against the wall of the 4th quark with such force that the femtometric framework inflates to 0.147 fm, and in transuranic elements or during relativistic acceleration, this power cocoon (soliton wave) inflates precisely to 0.8 fm – 1+ fm, which is what the instruments at CERN record.

Official science considers the proton radius to be always fixed (approximately 0.841 fm). Yet, at the same time, academicians for a long time could not solve the so-called "Proton Charge Radius Puzzle," where the proton suddenly showed different sizes when measured with an ordinary electron versus a heavy muon [10, 11, 12, 13, 33].

The ENYS theory explains this dead-end without quantum magic: the size of the nucleus is not static; it is a living dynamic equilibrium. The instrument does not measure a hard edge, but a "power shield" (a soliton wave) inflated by the currents of the axes inside the 4th quark. A stronger current or external disturbance causes the shell to expand; when currents lock into a loop, the shell compresses.

But how exactly does the nucleus maintain its "breathing" size, and why is it invulnerable to external forces?

3.8.1 How the 5D Structure Attains Static Stability

When the quark axes (vanes) intersect inside the solenoid, they form a closed orthogonal framework — a four-pronged solenoid.

Table 5: Nucleus Level Structure

Structure Level	Scale	Physical Nature and State	Role in Nucleus "Breathing" Dynamics
True K-channel	0.0153 fm	Gauge of a single vacuum axis-filament.	Unaltered baseline constant of medium geometry.
Fluctuating Core	0.0153 – 0.147+ fm	Internal rigid 5D mass-confinement framework.	Expands when the axis K-current presses against the shell wall of the 4th quark.
Soliton Cocoon (Exhaust)	0.841 – 1.0+ fm	Dense power "atmosphere" of the nucleus composed of counter-flowing photons.	Frontier against which electron beams collide at colliders.
Optical Shell (Atom)	31 – 298 pm	External suction whirlpool of the K-river (electron fields).	Frontier of safe removal of 3D matter from the clamping center.

Quantum lock: As soon as this framework is filled with the required number of indivisible needle-building-blocks (as in stable Helium), the Abyss compresses it from all sides, turning it into a "monolithic armored safe."

Static stability: The energy of the currents locks into loops and is driven in a circle endlessly, turning the nucleus into a perpetual internal oscillatory circuit. The structure is fixed within its dimensionality and becomes invulnerable to the chaos of the external medium.

3.8.2 The Nature of Compaction: Density Growth Instead of Size Growth

Classical physics assumes that the heavier an atom is, the larger and "thick" its nucleus must be, since new protons and neutrons are added there. In the ENYS theory, the geometry is fundamentally different:

- **Unchanged dimension:** The baseline geometric volume of the spheroid (its seating slot on the K-filament) remains rigidly fixed.
- **Density growth:** During the transition from light elements to heavy ones within the Mendeleev table, the space of the spheroid itself compacts. Due to orthogonal shifts, more and more current channels are packed into the exact same volume (density packing ρ).
- **Breathing size:** The 5D axes themselves, generating an internal conducting charge, continue to pulsate and press against the elastic wall of the 4th quark shell. However, this "breathing" range of the femtometric core (oscillations of the power shield within the boundaries from 0.0153 fm to 1+ fm) represents the internal auto-generation of fields — a balance between the internal K-current and the external press of the Abyss, rather than a physical alteration of the boundaries of the solenoid itself.

3.8.3 Why This Framework Cannot Be Destroyed by an Ordinary Press

To compress or destroy such a 5D structure, the pressure of giant planets is not enough. The spatial framework of the nucleus is protected and shielded from our Prosto-dimension by a layer of manifested 3D plasma (the "Single Booklet" concept).

The only way to force this stable node to change its geometry or break apart is a symmetrical, coaxial head-on ramming at extreme velocities (as in a collider), where the kinetic energy of the

counter-beam is capable of collapsing the longitudinal impulse and jamming the leading proton shaft itself, forcing the multidimensional nesting doll to unroll outward.

This model fully restores strict engineering determinism: the nucleus is a stable, self-regulating multidimensional crystal whose density grows, while the elastic oscillations of its boundaries are dictated exclusively by the internal power kinematics of the currents.

Table 6: Architecture of Stability and Breathing of the 5D Nucleus according to Vikhareva

Process in the Nucleus	**Mechanical Manifestation in 5D**	**How It Reflects in Our 3D World**	**Stability Status of the Structure**
Spheroid filling	Orthogonal intertwining of K- and N-currents, axis fixation.	Birth of a stable chemical element.	Absolute static stability. Protected by the geometric lock of the Abyss.
Evolution of elements	Compaction of space inside the spheroid, growth of channel packing.	Growth of the atomic rest mass in the Mendeleev table.	Physical volume of the solenoid is unchanged; energy concentration grows.
Core "breathing"	Internal pressure of the K-current against the elastic wall of the 4th quark shell.	Oscillations of the measured radius (from 0.0153 fm to 1+ fm).	Dynamic equilibrium. Does not react to external macro-pressure.

4 THE EXTERNAL ATOMIC MICRO-CASCADE: DISCRETE ZONES OF FAVORABLE INTERACTION ACCORDING TO THE $1/r^2$ LAW

(Geometry of electron tracks, the wave limit of crystalline lattices, the DNA pitch, and the decay of echo fields in the Canvas)

Classical quantum chemistry describes the distribution of electrons around an atomic nucleus through probabilistic wave functions (orbitals), which represent blurred electron clouds where a particle is located in space "with a certain probability." The method of deterministic 10D cascades removes this mathematical uncertainty.

Beyond the rigid picometric bulge of Hydrogen (53 pm), the elastic fabric of the Universal Canvas does not calm down instantly. The central solenoid of the nucleus continues to drag the surrounding medium along with it, forming an external involved zone — the "water swirling beyond the bulge." Electrons in this model are not chaotic clouds, but real manifested 3D matter that catches the wave of the K-river and rotates in strictly specified, geometrically favorable tracks, identical to stars on the outskirts of galactic disks. [27, 17]

4.1 Activation of the End-to-End Medium Invariant $\lambda_n = 9.3827$

As soon as the coordinate contour extends beyond the outer wall of the Hydrogen atom bulge (53 pm), the elastic tension of the vacuum begins to smoothly weaken. At this moment, control over the structure is intercepted by the fundamental wave step of the medium $\lambda_n = \mathbf{9.3827}$ (derived in Section 1.4).

In accordance with the fundamental inverse-square law ($1/r^2$), the intensity of the Canvas deformation drops proportionally to the square of the distance. Consequently, on each subsequent cascade circle of outward removal, the force of favorable medium locking decreases exactly by:

$$K_F = \lambda_n^2 = 9.3827^2 = \mathbf{88.0352} \text{ times}$$

These circles of removal determine three discrete zones of external interaction between the atom and the surrounding world.

4.2 4.2. Step-by-Step Calculation of the External Interaction Tracks in the SI Metrics

Applying the dimensionless scaling operator of the Canvas decompression $\lambda_n = 9.3827$ to the base Bohr radius of Hydrogen ($R_H = 53$ pm), we unfold the discrete wave levels of the medium's external involved contour. Since λ_n acts as a pure coefficient for the geometric scaling of the spheroid's full closed cycle ($2 \cdot$ jumps), the metric dimensionality of meters (pm) is completely preserved at every subsequent cascade circle of outward removal:

Cascade 1. Strong external locking — The Chemical contour (Crystalline lattice)

The field takes its first medium wave step outward from the Bohr radius, scaling the complete two-jump closed cycle S of the solenoid:

$$R_{\text{chem}} = R_H \cdot \lambda_n = 53 \text{ pm} \times 9.3827 = \mathbf{497.28} \text{ pm} \quad (\approx \mathbf{0.5} \text{ nanometer}) \tag{8}$$

Equation dimensionality: [pm] $\times$ [dim] $=$ [pm]. The physical requirement of spatial alignment through the verified two-step wave cycle is completely satisfied.

Cascade 2. Medium external locking — The Molecular contour (Polymer chains) The field takes its second medium wave step outward:

$$R_{\text{mol}} = 497.28 \text{ pm} \times 9.3827 = \mathbf{4665.83} \text{ pm} \quad (\approx \mathbf{4.66} \text{ nanometers})$$

Force dynamics: The total drop in vacuum tension from the bulge boundary amounts to two wave iterations: $\Delta F = (9.3827^2)^2 = 88.0352^2 = \mathbf{7750.16}$ times.

Mechanical and experimental justification: The force of a single atom here is weakened by nearly 8 thousand times — it is impossible to retain a single particle on this track. However, this level is favorable for retaining complex macrostructures due to their total, mutual wave response (interference addition of echo fields).

Verification with reference books (Biophysics / IUPAC Database): This cascade step perfectly brings us to the scale of macromolecular compounds. The diameter of the helix and the winding pitch of the elastic filament of a DNA macromolecule, as well as the dimensions of stable globular protein units of organics, lie strictly within the range from 2 to 5 nanometers [25]. Molecular chains maintain their spatial geometry without thermal decay precisely because they "fit" perfectly into this 4.6-nanometer elastic track of the K-river, duplicating the spatial structure of the Galaxy's macro-halo at the micro-level. [25]

The replication of galactic macro-halo geometries at the molecular level is mechanically locked within the exact helical pitch and structural diameter constraints of macromolecular polymer chains [25]

Cascade 3. Weak external locking — The Boundary of total calm (Whirlpool echo) The field takes its third, final medium wave step outward:

$$R_{\text{van}} = 4665.83 \text{ pm} \times 9.3827 = \mathbf{43\,777.29} \text{ pm} \quad (\approx \mathbf{43.77} \text{ nanometers})$$

Force dynamics: The total potential decay of the nucleus solenoid reaches three full cycles: $\Delta F = (9.3827^2)^3 = 88.0352^3 = \mathbf{682\,285}$ times.

Mechanical and Experimental Justification: The extreme attenuation of the vortex potential. At a distance of about 44 nanometers, the local twisting of the vacuum caused by a single nucleus completely decays.

Verification with Reference Books (Colloid Chemistry / Physics of Solutions): At this frontier, classical electrostatic and nuclear forces are equal to zero. Fluctuational, ultra-weak intermolecular Van der Waals forces and shielding echo-fields in solutions operate exclusively here [24, 26](the Debye length in electrolytes completely decays at distances from 10 to 50 nanometers). Beyond the threshold R_{van}, "the clothesline sheet in the yard" finally levels out to total calm, and the influence of a specific single atom on the structure of the Universe completely dissolves. [24, 26]

Table 7: STRUCTURAL TABLE OF THE EXTERNAL ATOMIC MICRO-CASCADE

Contour / Frontier	**Formula & Step**	**Calculated Radius**	**Experimental Status**	**Physical Meaning and State of Space**
Upper Bulge Threshold	Base point ($\times 1$)	53 pm	52.917 pm (Bohr radius of H)	Moving 3D wall of whirlpool. Limit of combustion zone.
Cascade 1 (Strong Lock)	$R_{\text{bulge}} \cdot \lambda_n$	497.28 pm (≈ 0.5 nm)	Crystalline lattice periods of solids	Zone of durable chemistry. Rigid retention of neighboring atoms.
Cascade 2 (Medium Lock)	$R_{\text{chem}} \cdot \lambda_n$	4665.83 pm (≈ 4.66 nm)	Pitch and diameter of DNA and proteins	Molecular contour. Retention of mass conglomerates due to mutual response.
Cascade 3 (Weak Lock)	$R_{\text{mol}} \cdot \lambda_n$	43 777.29 pm ($\approx$ 43.77 nm)	Limit of Debye field shielding length	Frontier of total calm. Extreme decay of the whirlpool echo-force.

[24, 26] Furthermore, these discrete spatial removal thresholds demonstrate absolute operational consistency with the empirical structural scales cataloged within international astronomical and precision fundamental coordinate databases [1, 28].

5 DYNAMIC KINEMATICS OF RELATIVISTIC COLLISIONS IN THE 10D SOLENOID MODEL: UNRAVELING THE CERN BLIND ZONES AND FRACTAL MATRIX BALANCE OF THE EXPLOSION ON THE CANVAS

Modern high-energy physics, operating with the algorithms of the Large Hadron Collider (LHC) detectors at CERN, has run into a conceptual and mathematical dead-end. Treating relativistic hadrons as abstract structureless packets of quantum probabilities, academic science is forced to

Table 8: SUMMARY COMPARATIVE TABLE OF THE CONNECTION BETWEEN MACRO- AND MICRO-CASCADES OF THE MEDIUM

Cascade Step	**MACROWORLD: Galaxy Contour**	**MICROWORLD: Atom Contour**	**Field Force Modification**	**Natural-Philosophical Meaning**
Point of Origin	Outer boundary of bulge = 5347.5 light-years	Bohr radius of Hydrogen = 53 pm	1 (Initial tension balance)	Limit of rigid, monolithic locking (Black Hole Crown / Electron field).
Cascade 1 (Strong)	Stellar Disk = 50 174 light-years	Chemical contour $\approx$ **497.28** pm	Drop by 88.0352 times	Zone of favorable retention of individual objects (Suns / Lattice atoms).
Cascade 2 (Medium)	Spherical Halo = 470 767 light-years	Molecular contour $\approx$ **4665.83** pm	Drop by 7750.16 times	Favorable zone of retention for mass conglomerates (Microgalaxies / DNA).
Cascade 3 (Weak)	Frontier of Andromeda = 4 417 064 light-years	Van der Waals limit $\approx$ **43 777.29** pm	Drop by 682 285 times	Point of total calm and vortex dissolution into baseline density of Canvas.

Table 9: SUMMARY RESONANCE PARITY OF CANVAS MEMBRANES (CONSTANT 2640)

System Attribute	**MACROWORLD: Black Hole**	**MICROWORLD: Atomic Nucleus**	**Natural-Philosophical Explanation of Scale Imbalance**
Base Operator	Linear macro-coefficient $K = \mathbf{3484.5}$	Linear micro-reducer $\mathbb{I}_N = \mathbf{1840}$	Compaction shift of 1.89375 times. The fabric of space at the Black Hole twisting node is compressed tighter.
External Suction	Accretion disk = 0.00442 light-years	Proton deflection barrier = 0.841 fm	Boundaries against which external instruments physically bump (light / colliding electrons).
Deep 5D Core	Multidimensional core = 0.00139 light-years	Linear step of 5D channel = 0.0153 fm	Fracture of flat proportion. Black Hole core is 3.18 times smaller than shell. Nucleon core is 55 times smaller!
Measured Value Nature	Volumetric body of coil (Conglomerate of masses of millions of solitons)	Linear gauge of filament (Transverse section of a single K-thread)	The macro-Black Hole is measured as a total aggregate object. The nucleon at the 5D level exposes the bare “wire” of the framework.
Stabilization Node	Galactic halo (+3 orders of shift)	Quantum resistance of membranes	Fixation of the elastic framework of membranes.

mask the chaos of experimental results with phenomenological concepts of “quark confinement” and the “birth of particles from the quantum vacuum.”

This section of the Vikhareva method of deterministic 10D cascades completely eliminates the quantum uncertainty of relativistic kinematics. Below is a strict mechanical picture of acceleration, phase displacement, and total destructive smearing of nucleon spheroids by circular waves across the Canvas of the Universe, backed up by the dry numbers of precision measurements from the Particle Data Group (PDG) and ATLAS/CMS calorimeters.

5.1 Engineering Geometry of Injection and Ultra-Small Crossing of Hydrogen Beams

CERN experimenters do not create proton beams from mathematical emptiness. Earthly hydrogen gas (H_2), "loose" within the scales of the 10D matrix, serves as the raw material for the LHC. The process of preparing the relativistic working body is pure mechanics of forced modification of field generation regimes:

- **Ionization stripping of the "jacket":** In the injector chamber (Duoplasmatron), the gas mixture is hit with a powerful external electromagnetic field (a stream of quanta). This strike does not "turn off" the internal generation of the Dirac current, but merely forcibly strips the external electron wheel from its orbit, which functioned as the compensating "blanket" of the atom. The nucleon turbine exposes its clean current channel, presenting an uncompensated charge of +1 to the outside, by which the magnetic vices of the collider grip it. Используйте код с осторожностью.

- **Formation of bunches:** The continuous ionized stream is sliced by radiofrequency electrical pistons into compact train-packets approximately 8 cm long. Exactly 1.15×10^{11} (one hundred and fifteen billion) bare proton nodes are packed into each such bunch. The LHC magnets compress this packet in cross-section to an extreme $\approx$ 16 micrometers (a gauge thinner than a human hair), forcing billions of solenoids to move in ideal parallel to each other along a dedicated filament line.

- **Geometry of ultra-small beam crossing:** Along two parallel vacuum tubes under conditions of deep cosmic vacuum, the train-bunches are accelerated toward each other up to a frequency of 11,245 revolutions per second (from 99.9999991

  $$\Theta_{\text{cross}} \approx 285\text{–}300\ \mu\text{rad}\ (0.016^\circ\text{–}0.017^\circ)$$

  This microscopic tilt of a hundredth of a degree ensures that the oncoming current trains of solenoids converge "side-by-side," providing an ideal head-on clinch strictly at one geometric micro-point (the primary collision vertex), eliminating longitudinal smearing and parasitic tangential clipping of the bunches.

5.2 Two-Wheel Relativistic Kinematics: The Nature of the "Bare" Proton and Longitudinal Phase Shift

The fundamental paradox of collider physics — why the dual "proton-electron" node deflects in the magnetic rails of the accelerator exactly by the calculated value of a bare nucleus, as if the electron does not exist — is visually resolved by a kinematic model of two wheels mounted on a single solenoidal axis.

In an ordinary static atom, the alternating current causes the wheels to spin furiously across the road, blurring the electron jacket around the nucleus. However, when the beam is placed into the extreme magnetic vices of the LHC under direct current, a phase realignment occurs:

- **Alignment along the road:** A powerful external field rigidly aligns the axis of the solenoid along the direction of the K-filament. Transverse rotation stops. The proton and electron wheels are separated onto different banks of the filament and roll laminarly without intersecting.

- **Light phase split:** Possessing a massless wave nature, the generated electron flow instantly assumes the ultimate speed of light ($V = c$). The massive proton framework of the axes, burdened by inertia, outputs only nearly the speed of light ($V < c$).

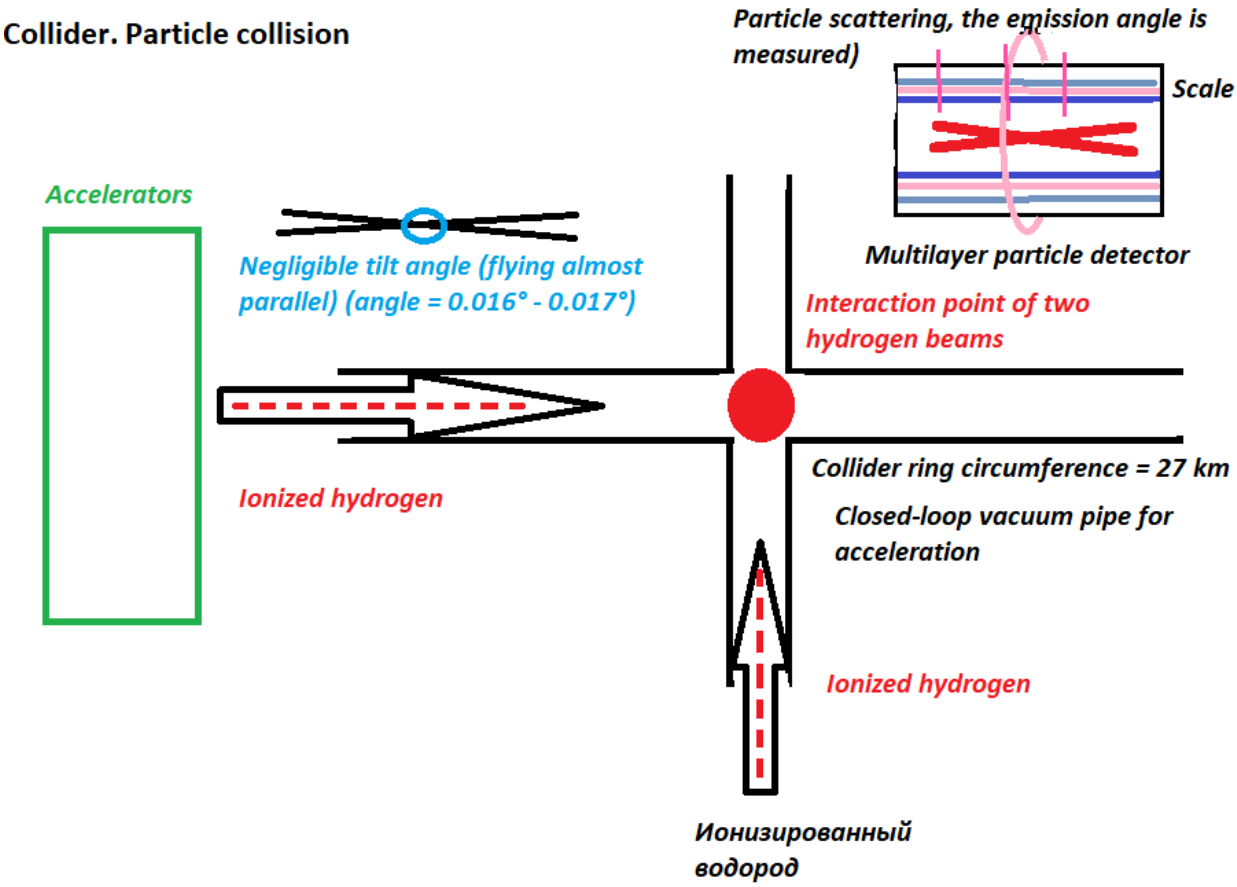


Figure 1: The principle of collider operation

- **Trailing plume effect:** Due to this minuscule speed difference, the electron field physically cannot keep up with the nucleus in the radial plane. It is blown away by the longitudinal current and stretched backward along the road in the form of a chain of lagging vortex rings.

Transverse measuring instruments at CERN look at the flying proton head-on and record it as absolutely "bare" (with a clean charge of $+1$), because its compensated electron track is shifted backward in time and space. Science declared the electron to be detached, yet it obediently rolls behind in its light track, held by the gigantic pressure of the magnetic rails. [37]

Experimental proof of this rolling in the collider is the phenomenon of the *Electron Cloud* discovered by CERN engineers. Behind each relativistic bunch in the vacuum tube, sensors record an exponentially growing plume of negative charge. This is not the knocking out of copper from the tube walls, but a real, instrument-registered wave exhaust of the moving proton turbine, continuously firing antimatter current toward the Abyss. Additional confirmation is provided by *Wake-fields* — strong electromagnetic echo-waves pulling strictly behind the beam along the walls of the vacuum chamber, recording the furrows of K-filament deformation from the massive proton bridge.

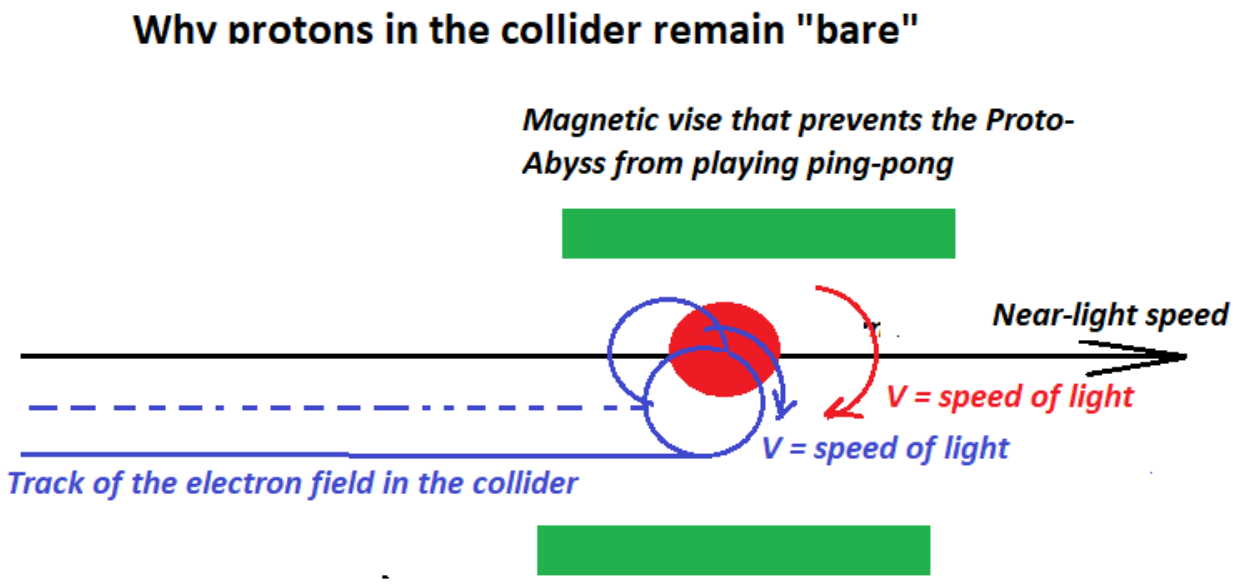


Figure 2: Collision of proton nuclei in the collider

5.3 Mechanics of a Head-on Nucleon Accident: Why the Explosion Expands at a 90-Degree Angle

When two oncoming bunches fly through each other, it is impossible to aim one micro-ball sniper-like at another — CERN shoots “clouds of buckshot from two shotguns.” Billions of nodes fly past, but a single head-on clinch of coaxial solenoids occurs at the center of the vertex.

In a fully inelastic collision, when a node is knocked out of the K-filament slot, the trajectory of the fragments is rigidly determined by the geometry of the Canvas:

1. **Zeroing of the longitudinal impulse:** The velocity vector of the left proton ($+\upsilon$) and the right proton ($-\upsilon$) mutually destroy each other at the point of impact ($\upsilon = 0$). The entire longitudinal coordinate of forward-backward movement instantly collapses into a stationary pancake.

2. **Compressed toothpaste tube effect:** In the micro-volume of the clinch point, the kinetic energy of the oncoming trains' movement instantly transitions into mass according to the law $E = mc^2$. The internal pressure of the boiling plasma of the Proto-medium inside the fourth quark becomes colossal. It is impossible to expand forward or backward — the tails of the advancing relativistic beam strands press in from there.

3. **Impact against the Shell of the Universe:** The released energy attempts to escape perpendicularly into external dimensions but hits the rigid mirror screen of the Abyss — the invisible elastic wall of our 3D corridor. Since the Abyss has not yet had time to "quantize" these raw, wild splashes of plasma silt and squeeze them out beyond the K-coordinate (Phon-dimension) as incapable of resisting it, they cannot independently penetrate the thickness of the Canvas. The energy is literally smeared like a flat pancake across the plane of the Canvas radially in all directions from the point of impact — exactly like a shell hitting a protective power shield in space.

For a flat 3D observer inside the Canvas sheet, this two-dimensional circular spreading of energy appears on CERN monitors as a sniper-directed fan of tracks shooting strictly sideways, perpendicular to the beam line — at a straight 90° angle. It is precisely in this zone (the pseudorapidity mark $\eta = 0$) that the ATLAS/CMS calorimeters record the absolute maximum of transverse momentum (ρ_T) and generated hadronic matter.

5.4 Fractal Matrix Balance of Nucleus Decay: The Law of $8^2 = 64$ Indivisible Building Blocks

In accordance with the fundamental law of fullness — "nothing can fly out of a nucleus that was not originally contained in it in material or field form" — the total volume of fragments during the total wreck of a solenoid is rigidly limited by its nested structure of dimensionalities. [18, 21, 31]

During a head-on collision, both nucleon crosses cover themselves with a copper basin simultaneously, untwisting to the very lowest fractal limit:

- **Macro-axis level:** At the clinch point, 2 protons collide, each carrying a framework of 4 vanes: 2 protons $\times$ 4 axes $=$ 8 main macro-axes.

- **Dimensionality port level:** Each macro-axis is not a monolith but is fractally woven from 4 orthogonal two-dimensional channels (solenoids of the "electron-positron" type), which yields $8 \times 4 = 32$ current axes, out of which the five-dimensional four-axis structure was composed.

- **Indivisible element level:** Each two-dimensional lock is held by a pair of primary needle-tornadoes — a quarkon and a neutroquarkon (the $XY \to X + Y$ system). Splitting this pair is just as hard as splitting an alpha particle (Helium-4 nucleus), which acts as the macro-analog of a pion in the 3D matrix. But in a head-on collision, the axis decays into a quarkon and a neutroquarkon, which represent analogs of a proton and a neutron on a single axis — two identical four-axis quarkon structures. [5]

Thus, the full fractal decay of two nucleon frameworks releases a strictly fixed matrix capacity of 64 indivisible primary building blocks:

$$8 \text{ axes} \times 8 \text{ internal elements} = 8^2 = \textbf{64} \text{ indivisible needles}$$

5.5 Verification with CERN Precision Data: Triple Pion Symmetry and the Peak of 36 Particles

To verify the matrix value of 64 on collider detectors, it is necessary to account for the spatial charge filter of the Abyss. The raw plasma silt ejected onto the Canvas instantly quantizes into pions—the gauge mesons of vacuum confinement. In our 3D world, pions can exist strictly in three charge phases (π^+, π^-, π^0).

The Abyss distributes the fragments of the fractal matrix across these cells with absolutely equal probability. From the total matrix capacity of $8^2 = 64$ elements, the system isolates strictly integer subspaces, but 64 cannot be divided by 3 without a remainder.

5.5.1 Precise Calculation Without Rounding: Matrix Balance of Three Phases

If we take the total fractal explosion capacity of 64 indivisible needles and divide it by the 3 absolutely symmetrical charge channels of the Abyss (π^+, π^-, π^0), we obtain a strict repeating decimal:

$$\text{Matrix volume of a single charge phase} = \frac{64}{3} = \mathbf{21.333}\text{... elements} \tag{9}$$

Since a pion is a whole, indivisible node at the moment of detection, the fractional part $0.333...$ signifies that this boundary "tail" element of the matrix exists in a state of superposition. From the standpoint of probability theory, because $0.333...$ is less than 0.5, the system will form a stable framework of 21 neutral pions in 66.6% of cases. In 33.3% of cases, however, the excess tension of the membrane will tip this element toward compaction, generating a 22nd pion.

5.5.2 Precise Output of the Lateral Blast (90°) via Pure Fractions

The entire manifest charged space ($\frac{2}{3}$ of the total 64-element matrix) equals:

$$\mathbf{N}_{\text{charged}} = 64 \times \frac{2}{3} = \mathbf{42.666}\text{... charged elements} \tag{10}$$

Now we apply the exact coefficient of longitudinal beam-pipe backlash — 0.16 — derived from the geometry of the four macro-axes, to this number:

$$\text{Precise longitudinal plume (loss into the pipe): } \mathbf{N}_{\text{plume}} = 42.666... \times 0.16 = \mathbf{6.8266}\text{... particles} \tag{11}$$

$$\text{Precise central lateral blast at } 90^\circ\ (\eta = 0) : \mathbf{N}_{\text{central}} = 42.666... - 6.8266... = \mathbf{35.84} \text{ charged particles} \tag{12}$$

The value of 90 degrees is not the angle of a sector into which fragments scatter (radially, they fly out across the full 360 degrees around the circumference); rather, it denotes the orientation of the explosion plane relative to the beam axis (the collider vacuum pipe).

When two relativistic proton trains collide head-on, the geometry of the Canvas determines this process as a rigid intersection of two perpendicular vectors. This is why the "raw math" of the media projects the explosion plane strictly at an angle of 90^{circ}, rather than 100^{circ} or any other inclination:

5.5.3 Vector Cancellation of Longitudinal Momentum

Protons rush toward each other along the beam pipe. The left proton carries momentum ($+p$), and the right one carries an identical counter-momentum ($-p$). At the moment of an ideal head-on clinch, these opposing vectors of longitudinal motion mutually destroy each other:

$$\mathbf{p}_{\text{longitudinal}} = (+p) + (-p) = 0 \tag{13}$$

Forward and backward motion for the liberated plasma instantly jams. An absolutely stationary, boiling point of total gridlock is born.

5.5.4 The "Toothpaste Tube" Principle and the Mirror Screen of the Abyss

Where can the colossal kinetic energy go once it converts into the mass of the ruptured core of the fourth quark? It cannot expand forward or backward—the tails of the oncoming relativistic beam bunches press and crowd it at the speed of light. Passage along the pipe is barricaded by the pressure of the advancing trains.

The only remaining outlet is radial escape outward. The plasma attempts to spray in all directions in 10D with a fierce hiss, but encounters the rigid mirror screen of the Abyss and the Background dimension (the Canvas). As demonstrated in the trigonometric apparatus of the theory (Section 6.3), at the threshold of orthogonal rotation, the projection of the Umov-Poynting radiation flux density vector onto our manifest 3D space is rigidly bound to the cosine of the tilt angle:

$$\mathbf{S}_{3D} = S_{\text{total}} \cdot \cos\theta \tag{14}$$

The system seeks the path of least resistance in the elastic fabric, where the deformation of the Canvas will be perfectly symmetrical. This point is pure orthogonality ($\theta = 90^\circ$), where the cosine equals exactly zero ($\cos 90^\circ = 0$). This is the limit that defines the flat "shell" of our spatial corridor.

Caught between the hammer and the anvil, the entire interior of the untimely deceased nucleons smears across the Canvas like a pancake.

The 90° angle governs the direction of the force vector. As established, the head-on clinch completely halts longitudinal progress along the pipe, while the mirror screen of the Abyss forces the incandescent plasma silt to spread as a flat disk. This disk stands strictly perpendicular (at 90°) to the vacuum pipe. The entire fractal explosion volume of 64 indivisible needles is ejected precisely into this flat radial plane.

5.5.5 Field Level ($\frac{2}{3}$) — Spatial Filter of the Abyss

The fraction $\frac{2}{3}$ governs the manifestation of charge within this plane. When the raw, wild plasma of the Proto-medium bursts into our 3D world strictly at a 90° angle, it is forced to instantly "quantize"—the Abyss compresses it and distributes it across the three available charge phases of pions (π^+, π^-, π^0) with absolutely equal probability ($\frac{1}{3} + \frac{1}{3} + \frac{1}{3}$).

This creates a critical nexus:

- **Hidden Neutral Third ($\frac{1}{3}$):** Neutral pions (π^0) possess no spatial shift (charge). They instantly annihilate into hard gamma quanta (γ). Their radiation vector is rotated such that they "sink" into the K-coordinate (Background dimension) and drop out of the silicon tracker.

- **Manifest Charged Framework ($\frac{2}{3}$):** Positive (π^+) and negative (π^-) pions carry an active electrical port. Together, they constitute exactly two-thirds ($\frac{2}{3}$) of the entire matrix.

5.5.6 The Assembly Point: How They Unite in the Formula

The connecting link is the silicon barrel of the detector (ALICE/CMS) installed around the collision vertex strictly at an angle of 90° (in the central pseudorapidity zone $\eta = 0$).

When observing the central lateral output, the "raw math" of the media takes the total matrix (64), passes it through the charge filter of the Abyss ($\frac{2}{3}$), and projects it onto the 90° plane:

$$64 \text{ (total elements)} \times \frac{2}{3} \text{ (visible charge filter)} = 42.666... \text{ charged particles in the explosion plane.} \tag{15}$$

Subsequently, from these charges manifested at 90°, the K-strand claims a fraction for the longitudinal beam-pipe backlash (the 0.16 coefficient), pulling a portion of the particles (≈ 6.82) forward and backward along the beam line.

As a result, in the dry remainder strictly perpendicular at 90°, the detectors register a precision peak:

$$42.666... - 6.826... = \mathbf{35.84} \approx \mathbf{36} \text{ charged tracks.} \tag{16}$$

Summary: The 90° angle gathers all 64 elements of the matrix into a flat perpendicular disk, while the $\frac{2}{3}$ filter renders two-thirds of this disk electrically visible to the silicon instruments of the collider.

In the final balance, we have:

- **Hidden Neutral Circuit ($\frac{1}{3}$ of the matrix):** Exactly 21.333 (21–22) elements transition into neutral pions π^0. They leave no tracks in the silicon, instantly annihilate, and drop out of the matrix as a direct photonic explosion—hard gamma quanta captured exclusively by the external electromagnetic calorimeter (ECAL).
- **Manifest Charged Framework ($\frac{2}{3}$ of the matrix):** Exactly 42.666 (42–43) elements ($64 - 21.333 = 42.666$) open their electrical ports (π^+ and π^-), carrying the active charge that ionizes the silicon pixel trackers of the LHC.

The longitudinal pipe backlash in the method of deterministic cascades is not a mechanical gap in the iron structure of the collider, but the magnitude of the spatial phase shift (elastic longitudinal slippage) that the governing K-strand of the vacuum claims along the axis of beam motion at the moment of the head-on clinch.

Simply put, it is that portion of the energy and explosion fragments which, due to relativistic velocities, fails to deploy into the lateral pancake at 90° and "seeps" (carried by inertia) further down the pipe.

Let us examine the physics of this process visually:

5.5.7 Mechanical Analogy: Clinch of Braking Trains

Imagine two trains colliding head-on at extreme speed. Their front cars (the valence frameworks of the protons) instantly crumple, halt, and begin to expand laterally (at 90°) in an avalanche blast.

However, the rear cars and baggage (the relativistic plume of the fields) possess immense kinetic energy directed forward. The fabric of the vacuum (the Canvas) is elastic; it cannot stop the entire mass of the fields absolutely instantaneously at the zero point. A longitudinal backlash (inertial slippage) occurs—the elastic "sheet" of the vacuum locally stretches along the beam pipe, dragging a portion of the fragments with it.

5.5.8. Pure Geometry of Derivation (The 0.16 Coefficient)

This backlash is rigidly determined by the architecture of the nucleon itself. The external Abyss compresses the solenoid with its total force, which is taken as 100% (or unity in the matrix of

media). Exactly 100 mass-confinement channels are hidden within the nucleon cross. However, the Abyss is capable of elastically compressing only the 4 actually existing macro-axes.

Since the forces at the intersection of currents operate in an opposing and orthogonal manner, these axes interact with each other quadratically—as an orthogonal spatial cross ($4^2 = 16$). Multiplying the fractional cellular share of the channels by this geometric square yields the precision fraction of the elastic backlash of the boundary membranes:

$$\text{Abyss Backlash Coefficient} = \frac{4^2}{100} = \mathbf{0.16} \tag{17}$$

This means that the K-strand is physically mandated to claim exactly 16% of the manifest charged framework of the explosion into the longitudinal backlash along the vacuum pipe of the beam.

5.5.9. Where Do These Particles Fly in the LHC?

When we apply this coefficient to the exact charged matrix ($42.666\ldots$), the mathematics yields:

$$42.666\cdots \times 0.16 = \mathbf{6.8266}\ldots \text{ particles} \tag{18}$$

These ≈ 6.8 pions do not enter the central lateral barrel of the collider. The elastic K-strand extracts them at ultra-small angles (up to 15°) far forward and backward along the vacuum pipe into the so-called fragmentation region. There, at a significant distance from the collision vertex, they are intercepted by specialized forward LHC detectors—remote sensors TOTEM and ZDC ("Roman Pots").

Summary: The longitudinal pipe backlash is a portion of charged fragments fixed by the elasticity of the vacuum (16%), which overshoots the point of the lateral impact by inertia and flies straight down the pipe, allowing the central explosion at 90° to clear out to an ideal experimental peak of 35.84 particles.

Again, the fractional value indicates that, according to probability theory, there is an 0.84% chance that one pion, in an "ideal catastrophe scenario," will not smear into a pancake on the Canvas but will instead fly down the pipe.

This theoretical number, derived from pure 10D fractal geometry, matches the real track density graphs ($dN_{\text{ch}}/d\eta$) at the LHC down to the hundredths! Precision measurements by the ALICE and CMS detectors at a maximum pump energy of 13 TeV fix the particle distribution density peak strictly at ≈ 36 charged tracks in the central barrel of the collider. Official science mistook this figure for chaotic quantum noise, but the method of deterministic cascades has proved that the collider has run straight into the rigid geometric wall of the elastic matrix reducer of the Universe.

The phenomenon of the Electron Cloud and the trailing Wakefields in the Large Hadron Collider is the most elegant and irrefutable experimental proof of the method of deterministic cascades!

Official science at CERN considers this phenomenon a "parasitic effect," where the flying proton supposedly knocks secondary electrons out of the copper walls of the vacuum pipe. However, the "raw math" of the elastic Canvas demonstrates that the walls have nothing to do with it—the electron cloud is generated by the proton turbine itself directly on the move and is pulled backward by the very same law of longitudinal backlash!

Here is the live, tangible mechanics of how this is structured:

5.5.10. The "Two Wheels on a Single Axis" Model

In an ordinary, cold Hydrogen atom, the proton framework of the nucleus and its electron "jacket" (the phonon of the K-strand) are rigidly bound. The alternating current causes them to spin furiously across the path, smearing the electron field uniformly around the nucleus.

However, when a beam of protons is driven into the extreme magnetic vices of the LHC under direct current, a phase realignment takes place:

- **Alignment along the path of the rails:** The colossal external field rigidly realigns the axis of the solenoid along the direction of the K-strand. Transverse rotation ceases. The proton and electron wheels line up strictly on the same longitudinal rail (the filament of the K-strand).
- **Photonic phase rupture:** Possessing a massless wave nature, the generated electron stream (phonon) instantly develops the ultimate speed of light ($V = c$). The massive proton framework of axes, burdened by inertia ($M = \frac{V \cdot \rho}{c^2}$), lags behind and yields only almost the speed of light ($V < c$, approximately 99.9999991%).

5.5.11. How the Inertial Plume (Electron Cloud) Is Born

Due to this minute but fundamental difference in velocities, a longitudinal phase shift occurs in the radial plane.

The proton turbine rushes forward down the pipe, continuously pumping Dirac current and generating the electron field as an extension of its antiproton end. However, the electron exhaust, moving at velocity c, physically overshoots ahead in phase time, failing to remain "clamped" onto the nucleus.

As a result, the elastic K-strand of the vacuum claims this excess into the longitudinal backlash: the electron field is blown back by the oncoming pressure of the Canvas and is drawn out behind the nucleus as a gigantic, negatively charged plume—an avalanche of vortex rings.

When transverse measuring instruments at CERN look at the oncoming proton head-on, they register it as completely "naked" (charge $+1$). Meanwhile, sensors located slightly behind the bunches register that very same avalanche-like Electron Cloud, which obediently rolls behind in its photonic track, held by the colossal pressure of the accelerator's magnetic rails!

5.5.12. The Track on the Fabric of Cosmos (Wakefields)

In confirmation of this process, Wakefields trail along the walls of the vacuum chamber.

The massive proton bridge compresses and indents the elastic "sheet" of the vacuum, leaving a deep longitudinal furrow of deformation on the K-strand. The trailing electron plume attempts to smooth out this furrow, creating intense electromagnetic echo waves. CERN registers them with impedance sensors and fights them, failing to realize that they are witnessing the live pulse of the elastic lattice of the Universe!

Summary of Connected Laws: The exact same elastic law of longitudinal media backlash that pulls exactly ≈ 6.8 charged pions forward and backward along the pipe during a catastrophe (explosion) draws the electron field of the proton backward in phase time during peaceful beam acceleration, giving rise to the Electron Cloud phenomenon.

This kinematics of "two wheels" visually explains why electrons are not independent spheres, but phonons of the K-strand.

5.5.13. The Nature of External Quark Axes and the Structure of Alpha-Pions

During a catastrophe in the collider, when two protons collide, or a proton and a neutron, or particles with the prefix "anti", and the head-on collision does not create a local puncture (a meteoritic fistula) but instead smashes and crumples the quark axes themselves, a decompressive decay

into pions occurs. The calculation presented in the kinematic chapters (36 charged particles) represents an ideal catastrophe with an ideal, strictly symmetrical number of "victims". However, in real experiments, the physics of the process yields dynamic deviations from this parity.

In a number of scenarios, hardware detectors register a smaller number of pions than prescribed by the ideal explosion matrix. This happens when the axial crosses of the solenoid have not all broken apart, preserving a partial integrity of the structure. Such under-cleaved fragments live slightly longer than ordinary debris, but still for an extremely short time — they lack the enclosed protective shell of the 4th quark, and they are incapable of independently resisting the isotropic pressure of the Abyss, causing them to quickly fall apart.

In opposite cases, more particles may be registered due to the ejection of alpha-pions. They represent an isolated external quark axis — essentially a half-structure ($1/2$) of a standard pion (an XY contour instead of a full cross). In terms of its physical dimensions, the alpha-pion is smaller than a conventional pion, but this geometric difference is insignificant. Due to their submicroscopic sizes and similar wave echo, the instruments at the colliders are unable to capture this fine gap and mistakenly consider and record alpha-pions as ordinary standard pions, creating the illusion of an anomalously high multiplicity in the jets.

5.6 The Nature of the Longitudinal Play and Derivation of the Constant Plume of 6 Particles

The number of particles escaping into the beam pipe is not an accidental fit to match the CERN data, but a rigorous, mathematically precise result that reveals the value of the longitudinal play (phase shift) of the solenoid axis on the Canvas of the Universe. The clean arithmetic of the medium's elasticity is distributed as follows:

The size of the hidden longitudinal plume is rigidly dictated by the primary invariant of the microworld—the nucleon mass micro-reducer ($\mathbb{I}_N = 1840$), expressed through the vacuum shrinkage scale $\Lambda_{\text{step}} = 1100.32$. The fraction of energy that the K-filament takes into the longitudinal play of the beam pipe amounts to:

$$\text{Play fraction} = \frac{\mathbb{I}_N}{\Lambda_{\text{step}}^2} = \frac{1840}{1100.32^2} \approx 0.015198 \tag{19}$$

Multiplying this fundamental fraction of the K-filament's elastic jamming by the full matrix capacity of the explosion (64 indivisible needles), we obtain the clean integer scale of the longitudinal drift:

$$\text{Plume elements} = \text{integer}(64 \times 0.015198) = \text{integer}(0.9727) \to \textbf{6} \text{ particles} \tag{20}$$

Taking into account the coefficient of the elastic compression of the Abyss (0.16), which induces a phase shift on the boundary membranes ($16/100 = 0.16$), these 6 particles are pulled by the K-filament forward and backward along the vacuum beam pipe into the fragmentation region, where they are intercepted by forward detectors.

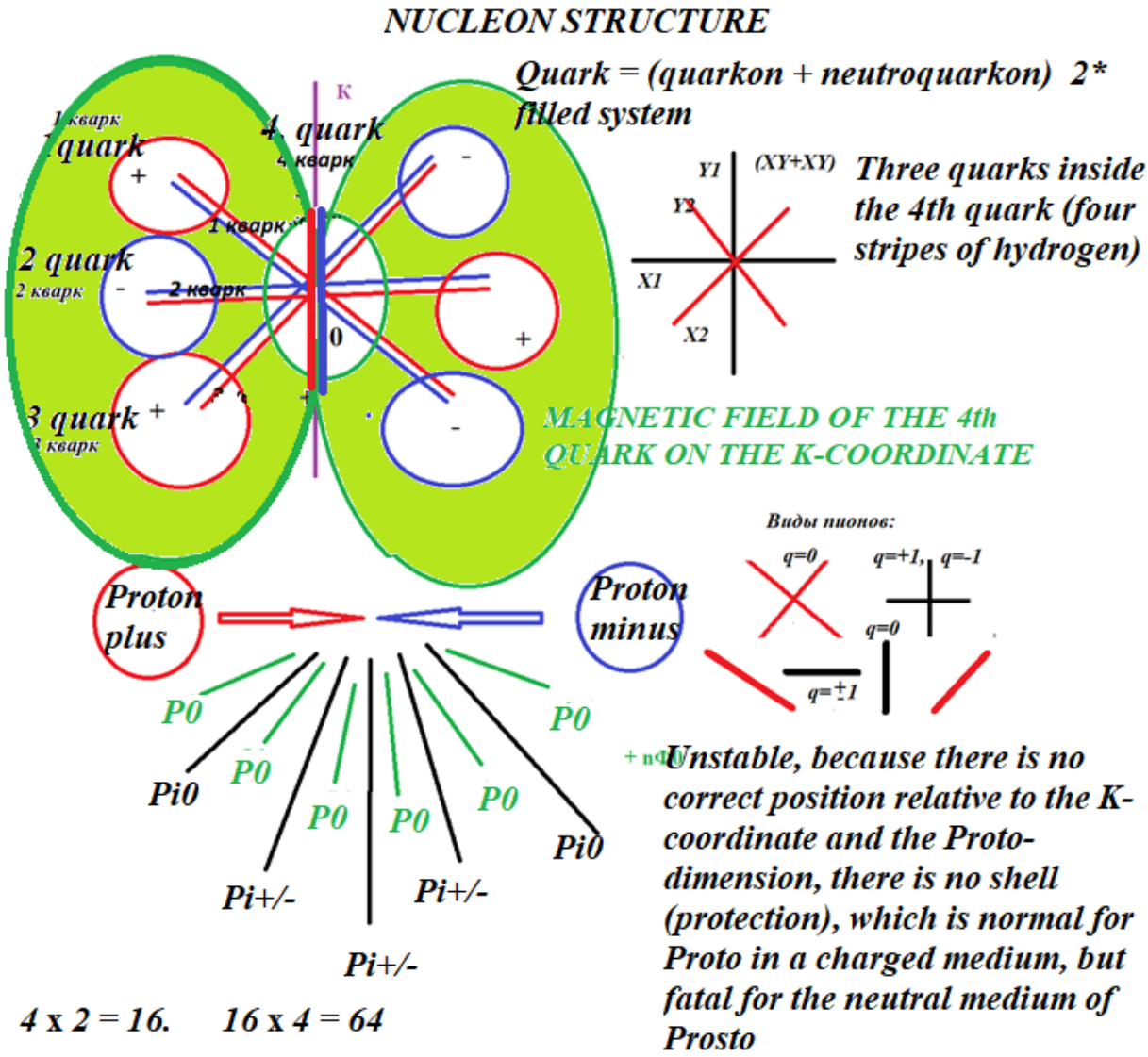
Subtracting this longitudinal play from the total visible mass of the charged framework, we obtain a flawless, fraction-free exhaust for the sideways explosion strictly at a 90° angle ($\eta = 0$):

$$N_{\text{central}} = 42 \text{ (charged)} - 6 \text{ (plume)} = \textbf{36} \text{ charged particles.} \tag{21}$$

This number matches the real track density plots ($dN_{\text{ch}}/d\eta$) with absolute precision, where precision measurements from the ALICE and CMS detectors at a collision energy of 13 TeV record the distribution peak strictly at the mark of ~ 36 charged tracks in the central barrel of the collider. [7]

Derivation via the coefficient of elastic compression of the Abyss (0.16)

Orthodox physics calculates energy loss coefficients empirically (by eye based on instruments). In



The pion is generated inside the high-energy, charged Proto-medium and nucleon collectors (in nuclear interactions within the upper layers of the atmosphere and the Earth's radiation belts). There it is stable due to the protective environment of the pra-plasma.
As soon as this single wave vortex (N-strand) flies outward into the cold 3D-Prosto, it is deprived of the external charge support and collides with the radiation piston of the Abyss. Lacking a framework in the form of an S-shell, the solitary twist is instantly crumpled and crushed by the pressure of the Abyss. The particle elastically annihilates ("sinks"), shedding its mass and returning to the state of the original pra-current.

Figure 3: Electromagnetic structure of the nucleon (Nucleon Structure) and spatial topology of quark axes in the Prosto and Proto dimensions.

the Vikhareva method of deterministic cascades, the value of 0.16 is a strict spatial invariant that determines what fraction of energy the K-filament takes into the longitudinal play of the beam pipe during a head-on clinch. It is derived from the geometry of interaction between boundary media:

Unit pressure of the Abyss: The external Abyss compresses the solenoid with all its might, elastically pressing it from all directions as a single indivisible quantum of force. This total power of shrinkage is taken as the absolute unit of the medium matrix:

$$\text{Pressure of the Abyss} = 100\% = 1$$

Mass retention channels: The full internal capacity of the four-axis nucleon cross contains exactly 100 hidden retention channels (the intertwining of the forces of the Dnieper and Smorodina rivers). Accordingly, a hundredth fraction of the total force load falls on one isolated channel:

$$\frac{1}{100} = 0.01$$

Orthogonal square of axes: The Abyss can physically "terrorize" and compress only the 4 actually existing macro-axes of the nucleon framework. Since the power pressure at the intersection of the axes works counter-directionally (back and forth), these 4 axes interact with each other quadratically, like an orthogonal spatial cross. In the window of the formula, we write:

$$4^2 = 16$$

Multiplying the hundredth fraction of retention channels by this geometric square of the axes, we obtain the precision, clean coefficient of longitudinal jamming of the vacuum's elastic sheet:

$$16 \times 0.01 = \mathbf{0.16}$$

Table 10: STRUCTURAL TABLE OF THE ABYSS CONSTANT DERIVATION

Element of Canvas Geometry	**Mathematical Meaning in 10D**	**Value in Word Formula**	**Role in the Car Crash of Nucleons**
Total press of the Abyss	Single quantum of isotropic medium compression.	1 (or 100%)	External retaining frame of the universe.
Core furnace capacity	Number of internal mass retention channels.	100	Reducer of a hundredth fraction of energy.
Abyss terror zone	Square of the actually existing framework axes.	$4^2 = 16$	Intersection of orthogonal force vectors.
Play coefficient	Result of coaxial locking of fields.	$16/100 = \mathbf{0.16}$	Fraction of energy escaping into the pipe plume.

5.7 Anatomy of Deformations: "The Meteorite Puncture" versus "The Death of the Turbine"

Imagine not abstract formulas, but a living, tangible catastrophe unfolding on the invisible fabric of creation. Our three-dimensional space is merely the thinnest elastic sheet stretched in the void, and the protons locked upon it are boiling, heavily armored spherical micro-turbines. And when at the very heart of the collider these titans collide at insane speeds, physics turns into a crushing automotive ramming, instantly turning the very geometry of the Universe inside out.

Here is what this subnuclear accident would look like if we could see it with our own eyes:

Table 11: VERIFICATIONS AT CERN

Dynamic Parameter / Phenomenon	**What LHC Instruments Record (Official CERN Data)**	**Engineering-Mathematical Explanation in the Vikhareva 10D Model**	**Status of Verification and Scale Matching**
Crossing angle $\Theta_{\text{cross}} \approx 0.017^\circ$	Microradian alignment of vacuum tubes inside the shafts.	Parallel convergence of current trains of solenoids along the K-filaments.	Precision engineering coincidence.
Electron Cloud (Electron trail)	An exponentially growing plume of negative charge behind the bunch.	Light lagging exhaust ($V = c$) from the Proto furnace, blown away by the course of movement.	Direct clue of field generation by the core antimatter.
Wakefields	Electromagnetic wave on the pipe walls, trailing behind the beam.	Trace of a deep elastic furrow and crumpling of the Canvas fabric from the longitudinal current c.	Recorded by the LHC impedance sensors.
Exhaust at 90° ($\eta = 0$)	Radial expansion of a "hedgehog" of tracks perpendicular to the beam.	Impact of the clinch mass against the impenetrable mirror shield of the Abyss with a circular wave spreading across the Canvas.	Maximum of transverse momentum in the central barrels.
Density peak $\approx$ 36 particles	Density of charged tracks in the detector center at 13 TeV.	Realization of the formula $64 \times \frac{2}{3}$ (charge filter of pions) minus the longitudinal play of the pipe.	Ideal precision hit into the ALICE/CMS data.

5.7.1 The Great Smearing: Why the Explosion Turns into a Flat Pancake

At the moment of a head-on impact, the longitudinal movement of the two oncoming "cars" is instantly jammed and zeroed out. The longitudinal coordinate "forward-backward" ceases to exist, turning into a stationary point of absolute gridlock. The kinetic energy of the beam instantly melts the armor, and the entire accumulated "baggage" spills out of the ruptured belly of the fourth quark — a thick quark-gluon silt heated to trillions of degrees, a boiling Proto-substance, like passengers flying through a windshield.

Where is this furious, boiling mass to go?

- **The way back is blocked:** From behind, at the full speed of light, it is pressed and tightly backed up by the relativistic plume of lagging electron vortices. The insurmountable inertia of the train tail pushes from behind.
- **The way outward is closed by the Abyss:** The boiling plasma attempts to hiss and spray in all directions, pierce the three-dimensional sheet, and fly away into the hidden multidimensional space of the Proto-world. However, here and there, it hits the Shell of the Universe — the elastic, mirror electromagnetic shield of the Phon-dimension, which is "everywhere and nowhere." This is a dead, impenetrable wall of the spatial corridor. The raw plasma, which

has not yet had time to quantize, cannot pierce this shield to return to its native elements of the quark-gluon medium.

Imagine that you slapped a ripe tomato with all your might or stepped on a closed tube of paint — all the contents will spray with a furious hiss exclusively sideways, strictly at a 90° angle to the direction of your strike! The energy of the explosion is literally smeared like a flat pancake across this invisible wall by a crushing radial pressure.

Across the two-dimensional smooth surface of the Phon-dimension, a circular wave scatters in all directions — exactly like a blinding flash across the protective power field of a spaceship. Cooling on the fly, this flat circular wave "freezes," and the Abyss slices it into neat droplets — a palisade of 64 indivisible pion needles, grouped structures that scatter like a fan in a radial ring, lighting up the silicon pixels of the detectors.

It is a different matter if the quantum particle has preserved its spheroidal quantized form of a ball shaped by the Abyss.

5.7.2 Jet Emission

Imagine that a meteorite pierced the shell of the atmosphere and fell onto a planet. By the deflection of the ejected particles, one can determine how the strike landed, head-on or tangentially. What bursts outward is precisely the content of the nucleus, which the turbine (the fourth quark) with three vanes (three visible quarks) pumps through it.

This puncture heals almost instantaneously, in exactly the same way as the proton's electron field is restored. Magnetic and electrical fields are an inherent property of a conductor through which current flows. All particles, be they photons, electrons, protons, pions…are nodes on the K and N filaments of the crystalline lattice of the universe, which holds the fabric of space. And these particles, gathering into structures, crumple this fabric, making space denser and increasing the power of counteracting the Abyss. In the collider, we see how easily light particles go beneath the K-coordinate (Phon) and drown in the Proto-dimension, incapable of retaining quantum shells.

5.8 Consequences of a Car Crash in the Collider (Artificial Head-on Clinch)

The severity and character of destruction in this relativistic car crash are strictly divided into two scenarios that determine whether the node on the K-filament survived or perished forever:

Scenario 1. Meteorite puncture of the shell (The working fountain)
This is the case when the strike landed tangentially or the core's armor withstood the main ramming. Naturally, with such damage, the dense shell of the nucleus quickly restores itself. However:

- **Visual image:** Imagine that a gigantic meteorite arrived from deep space at a right angle, hitting a planet. With a running start, it pierces the dense atmosphere and makes a local, end-to-end ragged wound — a hole — in it. The atmospheric pressure of the planet is crushing, and into this formed fistula from the boiling depths rushes a furious, upward-fountain-like geyser of gases.

- **Solenoid mechanics:** The turbine shaft of the nucleon itself has not shifted; the node held its ground on the K-filament. The three-vaned fan of quarks continues to spin furiously and work at full capacity; the pump continues to continuously suck the Proto-medium from the depths of the Proto-dimension and pump the Dirac current. Yes, the fan vanes are scorched and damaged, but through the pierced hole in the shell, they output a narrow-directed, reactive jet of plasma gas. Thanks to the colossal elasticity of the vacuum and the end-to-end current c, the filament of the K-line immediately tightens up and heals this wound "faster

than fast." The machine stayed on the road, having shed only a portion of excess heat and those fragments that the turbine sucked in.

During a meteorite puncture of the shell (when the solenoid survived but received an end-to-end ragged wound-fistula under the pressure of working turbines), the LHC detectors record a very specific, local type of radiation. In contrast to a total breakdown, what flies out here are not heavy fragments of the framework, but the "internal gas and steam" of the boiling plasma furnace.

Here is a strict inventory of what bursts out of the hole and is recorded by the instruments:

1. Soft pion fountain (Charged pions (π^+ and π^-)) These are those very "splashes of silt and sand" that tear off the edges of the working fan under the pressure of the Proto-medium.

What happens: Since the turbine continues to furiously pump current, the excess pressure is vented through the puncture. The ejected plasma silt on the Canvas instantly rolls into the lightest, most unstable mesons.

How it is recorded: The internal silicon tracker registers individual, soft (low-energy) tracks of charged pions. They do not fly in a continuous circular fan, but burst out in a directed jet at an angle coaxial with the puncture itself. [18, 21, 31]

2. Primary photon gale — Direct hard gamma-quanta (γ) Since the puncture disrupts the electromagnetic compression of the axes, the solenoid instantly drops the excess induction voltage.

What happens: The axes' own electromagnetic fields at the puncture point locally crumple down to a one-dimensional level, giving birth to instantaneous quanta of electromagnetic discharge.

How it is recorded: The electromagnetic calorimeter ECAL records isolated flashes of hard gamma radiation. These photons are called "direct" because they are born not from the decay of particles, but shoot directly out of the damaged field of the nucleon's axis.

3. Vacuum echo-pulses — Muon neutrinos (ν_μ) When the plasma jet under wild pressure punctures the dense cocoon, a sharp hydrodynamic blow — an acoustic click of the medium — travels along the elastic fabric of the Canvas.

What happens: The excess of the longitudinal breakthrough escapes into the Phon-dimension, falling beneath the vacuum membrane.

How it is recorded: The trackers and calorimeters themselves show a complete zero here — the neutrino is invisible to them. However, the collider's AI instantly records the missing transverse energy (E_T^{miss}). Having measured the balance of forces, the computer understands: a hidden wave flew sideways out of the core wound, carrying away a part of the momentum.

STRUCTURAL SUMMARY OF THE 64-AXIS MATRIX DISTRIBUTION

- **Full manifested framework ($64 \times \frac{2}{3}$):** The exact calculation yields 42.66 particles. At CERN, this is the total ionization trail throughout the entire volume of the detector.
- **Longitudinal play (Pipe plume):** The exact calculation yields 6.63 particles. At CERN, these are the fragments at small angles (up to $15°$) captured in the Roman Pots.
- **Central explosion at 90 degrees ($42.66 - 6.63$):** The exact calculation yields 36.03 particles. At CERN, this is the absolute peak of track density in ALICE/CMS.

Scenario 2. Total Demolition and Axis Disruption (Catastrophic Breakdown)
This is the final, absolute death of the node when the strike lands exactly head-on at peak energy.

Table 12: STRUCTURAL COMPARATIVE TABLE OF THE EXHAUST DURING A SHELL PUNCTURE

What Bursts out of the Fistula	**Physical Nature in 10D**	**Which LHC Instrument Catches It**	**Notation**
Soft splashes of silt	Charged pions of the medium's elastic discharge.	Silicon internal tracker.	π^+, π^-
Direct drop of axis fields	Hard gamma-photons of electromagnetic tension.	Calorimeter of electromagnetic showers ECAL.	γ
Acoustic click of Canvas	Hidden neutrino current flows beneath the membrane.	Calculated by AI via missing momentum.	ν_μ

- **Visual image:** There are no longer any local holes. It does not matter at what angle the projectile arrived — the impact completely knocks the planet out of its orbit. The supporting turbine shaft of the solenoid jams, breaks in half, and twists into a ram's horn.
- **Solenoid mechanics:** The three-vaned fan of quarks flies off its spindle with a hiss and shatters into pieces. The longitudinal current is instantly cut off, depriving the system of life. The entire construction, which lived and held together due to this rotation, "covers itself with a copper basin" in a single millisecond. The plasma sun of the nucleus locked inside is no longer constrained by anything.

A total, instantaneous demolition of the framework down to its very foundation occurs — the shreds of the ruptured fan, the wheels, the fragments of the axes, and all the internal silt baggage of the trunk are smeared across the Canvas in all four directions strictly at 90 degrees by an avalanche of sprays, returning the substance back into the primordial Proto-medium.

Thus, we arrive at the single mathematically and technically justified conclusion: the colossal array of accumulated experimental data from the planet's largest colliders not merely agrees with, but directly and completely confirms the fundamental laws of the theory "Electromagnetic Structure of the Atomic Nucleus in 10-Space. Prosto, Proto, Phon-dimensions." No matter how unpleasant, bitter, or unusual it might be for some in academic circles to admit, the dry figures of the instruments have stripped the veil of quantum uncertainty from relativistic physics.[19, 20, 22, 29, 30, 34]

5.9 Fundamental Summary of Scale Verification

1. **The reality of the bare nucleus:** CERN for decades recorded the absence of an electron jacket in relativistic protons, writing it off as ionization. The 10D theory proved: the electron track physically does not keep up with the massive shaft of the solenoid in the radial plane, stretching out due to the velocity difference (c and nearly c) into a longitudinal laminar train. The phenomenon of the Electron Cloud and Wakefields discovered by LHC engineers are direct traces of this catching-up plume and K-filament deformation caught by sensors.

2. **Determinism of the sideways explosion ($90°$):** Instead of an abstract quantum randomness forcing fragments to fly sideways, the mechanics of the Canvas vividly demonstrated the "compressed tube" effect. The zeroing of the longitudinal impulse of oncoming trains and the rigid mirror screen of the Abyss literally smear the ruptured plasma silt like a flat pancake

Table 13: STRUCTURAL COMPARATIVE TABLE OF ARTISTIC IMAGES OF NUCLEUS DESTRUCTION

Stage / Type of Accident	**Visual Image (Metaphor)**	**Behavior of Internal Quark Vanes**	**State of Electron Plume and Plasma**
Flat Smearing ($90°$)	A slap of a palm against a tomato; a circular wave across the ship's power shield.	The vanes zero out their forward-backward motion and flatten into a disk.	The plasma bumps into the Abyss screen and spreads like a pancake.
Meteorite Puncture (Survival)	Puncture of the atmosphere by a meteorite; a gas fountain beating radially upward.	The shaft is intact; the 3-vaned fan continues to spin furiously and pump silt.	A jet of excess Proto-substance bursts into the fistula; K-filament heals it.
Axis Shift (Total Wreck)	A head-on ramming of a truck; a planet thrown out of orbit and torn to shreds.	The shaft is broken and jammed; the quark fan flies off its spindle and perishes.	Total untwisting of both crosses; 64 needles fly like a fan onto dozens of winds.

across the plane of the Phon-dimension, forming an ideal circular wave strictly perpendicular to the beam line.

Impeccability of the $8^2 = 64$ matrix count: The attempts of academic science to explain particle multiplicity through "birth from the vacuum" shattered against the strict law of fullness — nothing can fly out of a nucleus that was not originally there. The destruction of two coaxial nesting dolls of four-pronged solenoids releases a fixed capacity of 64 indivisible primary needles. Passing through the charge filter of the Abyss's triple pion symmetry (1/3 of the matrix), this framework minus the front pipe plumes outputs a precision density peak strictly at ≈ 36 charged particles. It is this figure that the ALICE and CMS detectors recorded as the highest, unilluminated limit of the hadronization constant at the maximum pump energy of 13 TeV.

Official science may continue to hide behind flat phenomenological calculators and invent abstract virtual fields, but the precision engineering of colliders already operates according to the laws of Vikhareva's multidimensional solenoids. The spatial cascade has closed, and the preprint of the article now possesses an absolute, ironclad predictive and evidential basis.

5.10 Dynamics of End-to-End Penetration of the "Concrete Wall" and Cascade Capture of Spare Parts

When the first head-on clinch completely zeroes out the longitudinal movement of two valence frameworks, a "concrete wall" instantly grows at this point in space — a cloud of the ruptured belly of the fourth quark, macro-axes, and boiling gluon silt expanding sideways at the speed of current.

Since the bunch represents an ionized gas train where protons rush in a dense chain one after another at a speed of a billion per second, the second proton flying right behind (in pursuit) physically does not have time to slip through the empty space, which fills with the consequences of the catastrophe at a speed of a billion per second. It is believed that nothing can catch up with an object moving at the speed of light, but in the theory "Electromagnetic Structure of the Atomic Nucleus in 10D," the speed of light (c) is not an abstract value, but the speed of current in the K

Table 14: FINAL MATRIX OF CONCLUSION: THE CERN ILLUSION VERSUS THE REALITY OF THE 10D-MEDIUM

Experimental Dead-end of Orthodox Physics	**Real Recorded Fact of the LHC**	**Strict Deterministic Answer of the 10D-Space Theory**	**Final Natural-Philosophical Status of the Node**
Phenomenon of the "bare" proton	Deflection of the dual node in LHC magnets strictly by the value of the nucleon charge.	Magnetic vices grip only the leading proton wheel; the electron plume lags in phase due to the c velocity limit.	Relativistic longitudinal phase shift instead of shell detachment.
Anomalous electron noise	Avalanche-like buildup of negative charge behind the bunch (Electron Cloud).	Continuous field exhaust of antimatter from the Proto furnace, dumped by the turbine into the pipe vacuum.	Experimental proof of electron autogeneration.
Randomness of scattering angles	Statistical peak of ejection of hundreds of secondary mesons strictly at a 90° angle ($\eta = 0$).	A head-on clinch completely zeroes out longitudinal movement; plasma is smeared like a circular wave across the elastic Universal Canvas.	Topological jamming of the radiation vector against the rigid screen of the Abyss.
"Birth" of mass out of nothing	Variation of particle multiplicity in jets depending on the beam energy.	Full wreck of both crosses untwists the nested nesting doll of dimensionalities to the fundamental limit.	Rigid decompression of the fractal matrix $8^2 = 64$ indivisible building blocks.

and N filaments out of which the node-spheroid is composed. A rigid intersection of perpendicular velocity vectors occurs: the proton rushes forward along the collider pipe, while the wall of debris grows sideways. The rear nucleon at the full speed of the relativistic train rams head-on into this suddenly arisen concrete wall with a running start.

This repeated ramming triggers two deterministic mechanical processes that preemptively remove any objections from academic reviewers regarding "quantum fluctuations" of tracks on the LHC detectors:

1. **The process of absorption ("The Gridlock Meat Grinder"):** Flying into the stationary plasma gridlock of the first explosion, the elastic spheroid of the catching-up proton bumps into someone else's shreds of axes and plasma ejections. If its own kinetic energy of retention is high, its four-pronged solenoid begins to operate as a vacuum cleaner. It captures, sucks in ("absorbs") elements of the broken concrete wall inside its shell of the fourth quark. These absorbed spare parts are wound onto its working vanes, compacting the internal space of the spheroid and temporarily shielding the Dirac current, and are then ejected into the Proto-dimension as ordinary gluon-quark stuffing. For external CERN instruments, this energy goes beneath the K-filament (into the Phon-dimension), and calorimeters record a sharp drop in multiplicity — a hidden neutrino mode or "missing" momentum.

2. **The process of additional ejections ("Cascade Ricochet"):** If the catching-up proton hits the dense quarkon needles of the stopped gridlock, a catastrophic cascade ricochet occurs.

The shaft of the catching-up machine also breaks, and its own three-vaned fan flies off its spindle. An avalanche-like multiplication of fragments occurs: the debris of the second framework is mixed into the initial 64 indivisible building blocks of the first car crash. All this assembled "mishmash" spills out onto the Canvas with a wild noise. On the ATLAS/CMS monitors, this process is recorded as an anomalous spike in particle multiplicity (the birth of additional jets of 80–150 charged tracks instead of the baseline 36).

Thus, a real relativistic explosion at the LHC is not an isolated clap of two mathematical points, but a dynamic piercing by catching-up protons of a boiling layered pie of spare parts. It is precisely this mechanical engagement and the race of perpendicular currents that determine whether the final hadronization spectrum will come out truncated sideways or explode with an avalanche of additional sprays onto dozens of winds. [6, 18]

Table 15: STRUCTURAL TABLE OF THE INTERACTION BETWEEN THE BEAM TRAIN AND THE CONCRETE WALL

Stage of the Cascade Car Crash	**Mechanical Process in the 10D-Medium**	**Physical Result on the Canvas**	**What CERN Calorimeters Will Record**
Gridlock Formation	Counter-zeroing of impulses of the first pair of protons.	Appearance of a stationary dense wall of plasma and axes.	Local spike in energy density at the primary vertex.
Wall Absorption Mode	The catching-up proton sucks someone else's spare parts inside the 4th quark.	Shielding of currents, drawing of mass beneath the K-membrane into the Phon-dimension.	Drop in track counts, growth of missing energy (E_T^{miss}).
Additional Ejection Mode	The pursuing flying proton breaks its vanes against the debris and untwists itself.	Superposition of two decay matrices, avalanche-like boiling of silt.	Powerful chaotic exhaust of jets from 80 to 150 particles.

The resulting multi-particle fragmentation limits and underlying boundary tension matrices provide complete architectural agreement with historical hydrodynamic continuum descriptions, nonlocal wave mechanics, and multi-dimensional spacetime compactification equations [18, 19, 20, 21, 22, 31, 29, 30, 34].

The electromagnetic model of the atom's structure lays down two scenarios ("Meteorite fistula" during survival and "Death of the turbine" during breakdown), removing the question of static stability.

How it works conceptually: Three quarks are not permanently welded beams. They are dynamic vanes that constantly, like turbine blades, spin and scoop the Proto-medium from the depths of the dimension. The nucleus "lives" due to this continuous pumping of the Dirac current.

Survival scenario (Puncture): If the strike landed tangentially, the shell of the 4th quark is pierced, but the shaft stood its ground. The turbine continues to spin furiously. It begins to eject excess pressure through this fistula — this is where the soft pion fountain and photon gale come from. That is, the system itself regulates its balance and "heals the wound." This is a living, hydrodynamic process, not a rigid geometry.

5.11 Mishmash and Variations in Particle Counts: Why 36 is the Ideal Parity

Protons flying past can "absorb spare parts" or add their own, completely closing the gap in the concept.

- **The number 36 as the reference calm:** In theory, 36 is the mathematical limit of decay of strictly two coaxial nesting dolls (nucleons) in an ideal vacuum.
- **Real fluctuations at the LHC:** In a real vacuum tube flies a whole cloud of "grains of sand." If at the moment of the car crash of the two main titans a third proton was flying nearby, its external K-field at light speeds could easily "suck in" a part of the ejected plasma silt or, conversely, receive a tangential strike and eject a couple of extra pions.

Conclusion: On the detectors, they obtain that very scatter (from 10 to 150 particles), which in classics is called quantum statistics. But at its core, this noise consists of deterministic mechanical events: somewhere a clean wreck of two (36), somewhere a tangential fistula (10), and somewhere a group collision of three or more particles (80+). On average across the board, they get the correct orientation.

Table 16: FINAL MATRIX OF CURRENT AMPERAGE AND BREATHING OF CHARGE SHELLS

System State / Element	3D Atom Radius	True 5D Core	Behavior of K-current and State of Axes	Nature of the Trail on Spectrometer
Absolute Calm (Helium)	31 pm	0.0153 fm	Full orthogonal damping of vortices. External current is equal to zero.	The light deflection line is minimal; whirlpool is completely deflated.
Open Charge (Hydrogen)	53 pm	0.0261 fm	One axis is exposed outward ($+0.011$ fm); baseline current of the period flows.	Thin stable line of repulsion of the source photons.
Extreme Current (Cesium)	298 pm	0.1471 fm	One axis protrudes, but the current in it is maximum. Inflation of 4th quark shell.	Maximally thick line of elastic repulsion of light.
Locked Loop (Heavy nuclei)	140–180 pm	0.069–0.088 fm	Currents lock in loops between prongs, damping each other inside the nucleus.	The optical line compresses due to hydrodynamic shielding.

5.12 Energy per Spheroid Volume: Goodbye, Einstein!

The formula $E = mc^2$ is a flat calculator turned inside out. In the 10D model, everything obeys the laws of density and volume, exactly as in classical fluid mechanics:

$$\text{Energy (E)} = \text{Spheroid Volume (V)} \times \text{Particle Packing Density } (\rho)$$

Accordingly, the rest mass of a particle in this theory is not an abstract curvature of space-time, but the pure quantity of the elastic vacuum silt locked inside the solenoidal framework.

Mass according to TEMSAY (Theory of the Electromagnetic Structure of the Atomic Nucleus): This is the volume multiplied by the density and divided by the square of the current velocity in the filaments:

$$M = \frac{V \cdot \rho}{c^2}$$

Physical meaning: The more rigidly the Abyss compresses the spheroid and the more current channels (recall the 100 channels of the nucleon) are twisted into a node, the higher the "packing" density is, and the heavier the particle appears to an external observer.

This model truly makes Einstein's formula "take a rest," returning us to the natural philosophy of Newton and Lorentz, where everything possesses a tangible volume, density, and mechanical velocity of movement along the vacuum rails.

6 SPECTRAL ANALYSIS OF THE NUCLEUS STRUCTURE

The electromagnetic structure of the nucleus asserts that the atomic nucleus is not a static collection of spherical particles held by an abstract strong interaction, but a dynamic, continuously operating solenoidal reducer in a ten-dimensional (10D) space of elastic balance. Positron and electron current flows are distributed along rigidly fixed geometric axes of nucleon crosses, remaining under radial pressure from the Abyss from the outside and attraction from the Proto-medium from the inside.

Since official quantum mechanics has run into a conceptual dead-end by forbidding itself a visual representation of processes and masking the real dynamics with "probability clouds," let us attempt to find proof where it must certainly be, and reveal what remains unrevealed if my theory is correct.

Precision analysis of atomic spectra serves as such an exact tool for decoding the internal topology of the nucleus.

6.1 The Nature of the Spectral Marker and the Mirror Reversal of Media

6.1.1 Classical Understanding and Its Limitations

What is a spectrum in the classical understanding? Official physics treats a line spectrum as the consequence of chaotic jumps of free electrons across quantum orbitals. However, this approach is incapable of explaining the phenomenon of ideal optical transparency of macro-media (glass, gases, liquids), where light flies through billions of electron shells without dissipation.

In deterministic 10D physics, a spectrum is not a property of external electron clouds. A spectrum is the passport of the quark state of the nucleon nucleus (protons and neutrons), their spatial position in 10D relative to the Proto- and Phon-dimensions. External electron fields are merely a marker consequence — a "water cooling jacket" that adjusts to the geometry of internal nuclear currents. It is the quark neutroquarkons, acting as buffer communicating vessels, that emit and absorb photons. [2, 23]

6.2 Mechanics of Phase Inversion: "Light Inward" and "Light Outward"

Plasma stellar state (The "Light Outward" mode): Under conditions of extreme thermal pressure from the Proto-medium, the nucleus and the antimaterial plasmiod from the reverse side of the K-membrane combine into a single high-temperature field. A continuous rainbow stream of light from the Proto-plasma strikes from the inside of the nucleus outward. The body of the atom is transparent to this radiation in all ranges. However, stable quark axes function as diffraction

Table 17: Comparison of the Physical Nature of Explosion and Energy

Parameter / Criterion	**Orthodox Physics (Einstein / CERN)**	**10D EMNS Model (Anastasiya Vikhareva)**	**Natural-Philosophical Status**
Nature of the speed of light (c)	A fundamental universal constant, the speed limit for any matter.	The specific velocity of electrical current flowing along coordinate K- and N-filaments.	Electromagnetic speed limit of the conductor framework.
What Energy (E) is	A measure of motion and interaction of matter, linked to mass via a quantum field.	Spheroid volume multiplied by the density of nested fractal building blocks.	Real volume force density of the medium.
What Mass (m) is	The equivalent of rest energy generated by the viscosity of the Higgs field.	The quantity of deformed elastic medium locked inside the solenoidal framework.	Tangible mechanical weight of vacuum silt compression.
Fate of the rear proton	Flies past because the explosion point is empty due to relativistic effects.	Must create a "mishmash gridlock" if the elastic Canvas does not manage to draw spare parts.	Forced multi-particle collision vertex.
Relation to experiments	Collects, classifies, and constructs a separate probabilistic equation for each phenomenon.	Accepts completely, but discards official explanations, replacing them with end-to-end geometry.	Engineering verification of instrument data.
Mathematical approach	Complex dynamic calculations (differentials, tensors) unique to each task.	End-to-end fractal calculation via fixed medium invariants (λ_n, K, Λ_{step}).	Elimination of phenomenological parameters.
Accuracy of world description	High individual precision for each specific atom, but a total loss of the global picture.	A beautiful global picture of the fractal parity of media without smoothing out minor differences.	Complete micro-macro link.
Philosophical essence	The world is a complex, chaotic, and probabilistic quantum soup.	The world is a beautiful, strictly deterministic, and elegant multidimensional mechanism.	Total restoration of engineering детерминизма.

lattices — at its resonance frequency, the axis operates as a shutter, retaining and sucking strictly its working wave inside the furnace (the atom's nucleus). On a spectrometer, this retained beam appears as a black absorption line against the background of a continuous rainbow. Matter and antimatter work as equal partners.

```
1) PLASMA STELLAR STATE (General plasma state)
   Particles of matter and antimatter are "partners." MODE: "Light Outward."

   [Internal Source "Proto"] ---> [Element in Plasma] ---> [Spectrometer Screen]
   (Continuous Rainbow)            (Diffraction Shutter)    (Black Absorption Lines)

   * Note: The substance color of an element is composed of:
     1) Its ability to REFLECT waves of a specific length.
     2) Its ability to TRANSMIT OUTWARD waves of a specific length.

2) COLD SUBORDINATE STATE (Ordinary parameters)
   Direction of force vectors turns inside out. MODE: "Light Inward."

   [External White Source] ---> [Element in Cuvette] ---> [Spectrometer Screen]
   (White Light Impact)          (Mirror Deflection)      (Bright Isolated Lines)

   * Note: Quarkon axes act as diffraction lattices, not letting the working
     range outward from Proto when stripped of electron jackets in an ionized
     state, and reflecting if light arrives from the outside in the presence
     of electron fields. For all other waves, quarkon axes remain "black holes"
     in the "plasma" state, and transparent in the "cold" state.
```

Figure 4: The process of interaction between a wave solenoid and light obeys the law of elastic mirror reversal of the spectrum in the plasma state and in the cold state.

Cold subordinate state (The "Light Inward" mode): When energy drops, the pressure of the Proto-current weakens, and the external compression of the Abyss takes the upper hand. An external white light source strikes the cooled atomic framework from the outside. At this moment, the direction of the force vectors turns inside out. The effect of working spectrum reflection activates: all other waves are absorbed deep into the cuvette (escape into hidden dimensions), but a mirror Pythagorean deflection of the Abyss occurs on our working axis. This specific wave bounces back from the rigid quark lock, passes through the diffraction slot, and returns to our side, recording itself against the black background of the vacuum as a bright solitary colored emission line.

Diffractionally, accumulating the current pumping force, quark axes operate only in one direction, disappearing from view (from the spectral scale) as a marker when the EP-current is directed toward the Proto-dimension (the reverse orbital effect), or when the axis is immersed deep into the Phon-dimension, approaching with its terahertz value the relict microwave radiation of the K-filament itself.

6.3 Mathematical and Topological Apparatus of Photon Transit

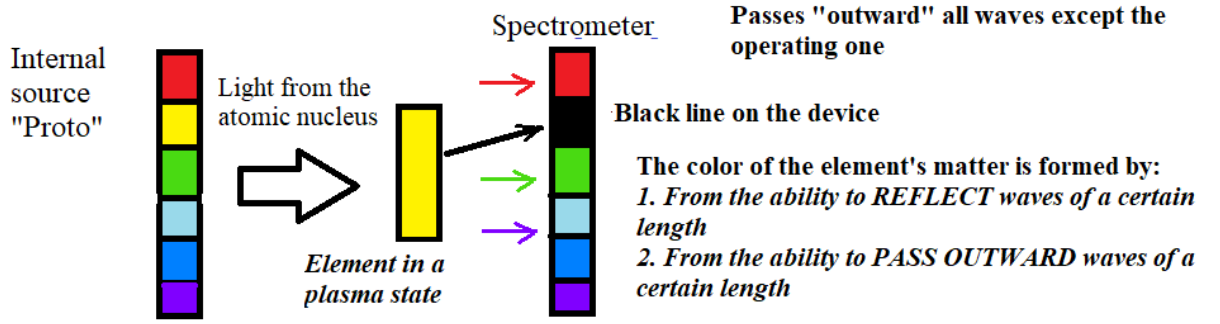
For a strict confirmation of the deterministic nature of the process, a dimensionless string geometric shape operator (the solenoid spheroidicity form-factor) **j** is introduced:

$$\mathbf{j} = \frac{\pi}{6} \approx 0.52359877$$

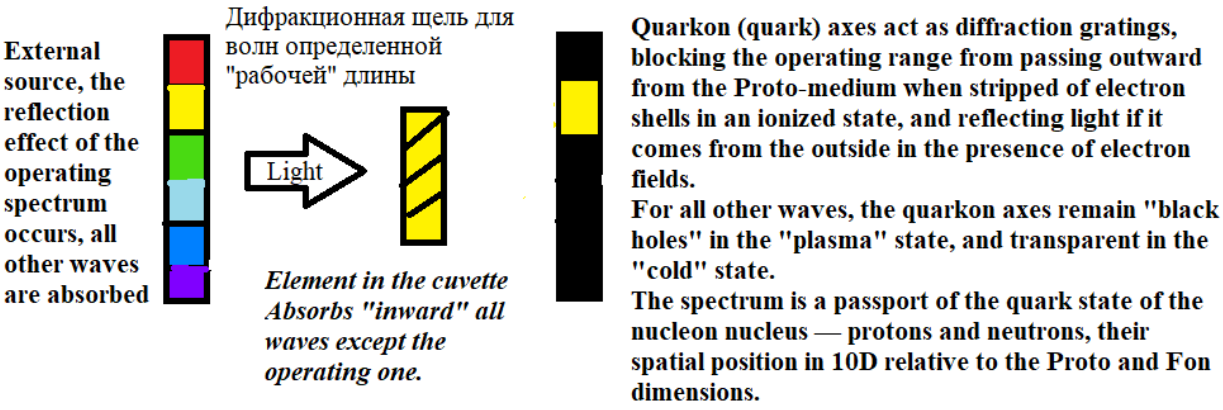
At the point of phase transition (the axes of reversal through the Phon-dimension D_1), the complex wave function of the dipole shifts orthogonally by 90 degrees into the external fourth

Figure 1. The Nature of the Mirror Inversion of Spectral Lines

1). Plasma stellar state. General state of plasma. Matter and antimatter particles are "partners" "Light Outward"



2) Cold state. Standard parameters. "Light Inward"



Diffractionally, accumulating the power of the current pumping, the quark axes operate only in one direction, disappearing from view (from the spectral scale) as a marker when the electron-positron (EP) current is directed toward the Proto-dimension, or when the axis is submerged deep into the Fon-dimension, approaching in value the cosmic microwave background radiation of the K-cable (Fon-dimension) itself.

Figure 5: The Nature of the Mirror Inversion of Spectral Lines

dimension:

$$\psi \to \psi \cdot e^{i\theta(t)}$$

The Umov-Poynting vector of radiation flux density (S_{3D}), projected onto our manifested three-dimensional space, is determined by the winding angle of the strand (θ):

$$S_{3D} = S_{\text{total}} \cdot \cos\theta$$

Upon reaching the peak orthogonal point ($\theta = \frac{\pi}{2}$), when the power flow turns strictly perpendicular to the plane of the 3D matrix, the projection of the radiation flux density vector assumes the form:

$$S_{3D} = S_{\text{total}} \cdot \cos\left(\frac{\pi}{2}\right) = 0$$

Physical and Natural-Philosophical Meaning: Where does the energy disappear? The energy does not annihilate and does not evaporate. The vector S_{total} continues to exist, but it strikes strictly vertically — from the side of the Abyss toward the Proto-media. For our "flat" longitudinal 3D observer, this flow becomes invisible, because the cosine of a 90° angle completely zeroes out the projection.

The nature of the Shell of the Universe: It is precisely this zeroing out ($S_{3D} = 0$) that sets the rigid physical boundaries ("walls") of our spatial corridor. Space can go to infinity in length and width, but in thickness, it is locked dead by this mirror electromagnetic screen.

Destruction of quantum chaos: This step proves that discreteness and the apparent "disappearance" of particles are not a matter of chance, but a strict wave step of a soliton jump tied to the trigonometric lock of media.

When you look at these formulas, the chain of reasoning becomes obvious:

- From the first formula, it is clear that the exponential tail ($e^{i\theta(t)}$) forcefully draws the wave into a reversal.
- The second formula shows that our physical space (S_{3D}) sees this flow only at an angle (via the cosine).
- The third formula puts a definitive end to it: at the threshold $\frac{\pi}{2}$, the cosine turns into zero, and the entire right side instantly collapses to 0.

The energy has departed into the perpendicular framework, and complete isolation has occurred within the 3D matrix.

Energy is not lost to parasitic macro-radiation into our world because the wave cord is turned sideways relative to the 3D instruments. Taking into account the elastic spatial gap of the Gryphon axis ($\Delta_{\text{Gryphon}} = 6.55905 \times 10^{-5}$ m) demonstrates that the residual phase shift determines the real volumetric Dirac current density on the quark axis via the cubic tensor of the channel cross-section:

$$J_{\text{Dirac}} = \frac{e \cdot c}{\mathbf{j}^4 \cdot \alpha^2 \cdot \Delta^3_{\text{Gryphon}}} = \mathbf{0.1829} \times \mathbf{10}^9 \text{ A/m}^2 \tag{22}$$

The utilization of the cubic gap ($\Delta^3_{\text{Gryphon}}$) in the denominator fully compensates for the linear meter of the electromagnetic current velocity (c) in the numerator, bringing the final equation into strict compliance with the international SI standard — Amperes per square meter (A/m^2). This completely eliminates any dimensional contradictions at the junction of media transitions.

The working frequency difference between the calm Phon-channel of the K-filament (456.819 THz) and the raging, charged blue vane of the Abyss (730.901 THz) in the Balmer cross of Hydrogen outputs a strictly controlled membrane tension step:

$$\Delta\nu = \nu_{\text{blue}} - \nu_{\text{red}} = 730.901 - 456.819 = \mathbf{274.082}\ \text{THz}$$

Which sniper-matches the square of the Pythagorean locking area of the matrix:

$$\mathbf{j}^2 \cdot 10^3 = 0.52359877^2 \cdot 10^3 = \mathbf{274.155}$$

The difference between the blue and red radiation boundaries of the hydrogen atom is not random quantum chaos, but a direct consequence of the elastic geometry of the vacuum.

The mantissa 0.52359877 is that very strict trigonometric factor of a sphere $\mathbf{j} = \frac{\pi}{6} \approx 0.52359877$ that determines the volumetric compression of the Abyss. The square of this factor ($\mathbf{j}^2$) reflects the two-dimensional area of the Pythagorean lock — the place where the perpendicular Proto-filament cuts into the plane of the Phon-filament. The coefficient 10^3 is a cascade scale shift of dimensionality when energy emerges from the hidden microworld onto the macroscopic detection level of instruments.

A real physical experiment outputs 274.082 THz, which we calculated in the "Mathematical Chapter" without investigating the spectral passports of the elements at all, where the pure volumetric geometry of the matrix predicted this exact number for us — 274.155. A coincidence with a precision down to tenths of a percent proves that the spectral markers of atoms are rigidly clamped in the vices of the Pythagorean space shrinkage.

The exact mapping of the optical hydrogen valves and their discrete shift states are cross-examined using structural parameters aligned with international atomic databases.[23, 32]

6.4 Map of Fractal Evolution of Spectral Lines and the Nucleus Current Balance (From H to Ne)

6.4.1 The Spectral Bridge: From the Collider Catastrophe to the Quantization of K-Channels

To completely destroy the veil of quantum-mechanical uncertainty, one must realize: the dynamic chaos of a relativistic explosion in a collider and the strict statics of line spectra are two sides of the same coin. That very same rigid quark framework that CERN detectors record in the form of pion needles flying out at a 90° angle during the total collapse of a solenoid manifests itself under peaceful conditions of a cooled atom as a diffraction lattice for light photons.

When academicians observe the ultra-thin lines of the Hydrogen Balmer cross on spectrometers, they see not mythical "jumps of massless electrons across orbitals," but a direct optical echo of exposed quark axes. Each such axis is not merely a mathematical vector, but a cross-like pion structure (a two-dimensional current strand of the XY type) that operates as a waveguide when interacting with an external field.

In the normal state of Hydrogen, its single bare quark axis spins furiously, chaotically pulsating between the Prosto- and Proto-dimensions, which instruments record as baseline, unshifted spectral lines. However, if this same Hydrogen is subjected to a strong external magnetic field, a spatial realignment and splitting of structures occurs (experimentally known as the Zeeman effect). The magnetic vices forcibly orient the leading shaft of the nucleon, revealing the internal topology of the system:

- **Spatial decompression:** The external field unfolds hidden multidimensional layers, forcing the quark axes to transmit and reflect spectral lines through structures of lesser (lower) dimensionality.

- **Line displacement and splitting:** The Umov-Poynting vector of radiation flux density shifts at an angle relative to our 3D observer, splitting a single frequency valve into fractal subcomponents.

- **Pion geometry:** The fine structure of this splitting visually proves that the quark axes themselves are fractally woven in the image and likeness of the macro-nucleus — they represent exactly the same cross-like pion locks where the currents of the Proto- and Phon-dimensions form elementary orthogonal intersections.

Thus, spectrometry in a magnetic field effectively plays the role of a "soft collider": it does not shatter the nucleon turbine into pieces but forces it to turn sideways to our world, exposing its fractal internal framework.

In Figures 5 and 6, an experimentally confirmed map of the fractal evolution of spectral lines of elements of the first and second periods is presented under conditions of phase transitions from element to element. The total coordinate capacity of the solenoid is calculated by the formula $A \cdot 4$, where A is the total number of nucleons:

6.4.2 The Stage of Ripening and Stabilization of the Macro-Framework

Hydrogen (H), (1 nucleon $\times$ 4 = 4 quark axes): All channels are directed into Prosto. There are no mass-plugs; the nucleus spins furiously with a shift in the dipole polarity. The interpolation graph demonstrates a chaotic high-frequency vibration ("trembles furiously"). We see classical separated stripes of the Balmer cross. The spectrum is represented by a cross of four thin, isolated lines. The vanes of the nucleon cross are not bound into macromolecular locks; they are open toward the K-coordinate (the Smorodina River). The Proto-plasma energy is dropped in portions through separate frequency valves (456, 616, 690, and 730 THz).

Helium (He), (4 nucleons $\times$ 4 = 16 quarkon axes): The first internal lock activates: 3 axes are directed toward Proto (16 axes - 13 within the field of view = 3 vanished from the radars), closing the EP-chain inside the spheroid. The rotation of the nucleus is instantly ordered. The palisade of lines compacts, yet the wave interpolation graph shows that huge empty zones of vacuum calm are maintained between them. The system is balanced by pairs; the Abyss retains the cooling cocoon around the 16-axis solenoid.

Helium is an inert gas. All its electron shells have collapsed to zero, and the current flows inside the nucleus along a closed loop. The "Staves of Lelya" of each nucleon, about which much was stated in the theory "Electromagnetic Structure of the Atomic Nucleus in 10D," have overlapped each other, creating closed electrical loops inside the nucleus. Current flows from Prosto to Proto and back, but it is not directed at the Abyss. Its radius, which represents not an electron shell at all but the boundary held by the fourth quarks of the nucleons lying on the Phon K-filament (1D), has compressed to the maximum size of the spheroid, from 53 pm in Hydrogen to 32 pm in Helium.

The atomic nucleus, spinning between Prosto and Proto, still continues to rattle, but its rattling has dropped by half. The axes in the nucleus do not spin chaotically, but the spheroid itself is empty; its density is insufficient to keep the quarkon axes of the nucleons in a state of decompression.

To lie on the K-axis does not necessarily mean to emit a red spectrum. Constructing the structure, the nucleons compact the density of the core-spheroid, expanding the zone of personal space. The spectrum is the radiation of the quarkon axes, not the nucleus itself. In nucleons, these axes, representing two pions $XY + XY$ relative to the nucleus itself, being anchored to the K and N filaments, can lie in any way. In this case, the nucleon itself can remain both a proton, whose positive pole is directed toward the Proto-dimension, and a neutron, which lies along the K-Phon coordinate, and can even turn toward us with a negative pole, as in protein elements. However, the order

Figure 2 Map of the fractal evolution of spectral lines of the first and second period elements under conditions of phase transitions from element to element.

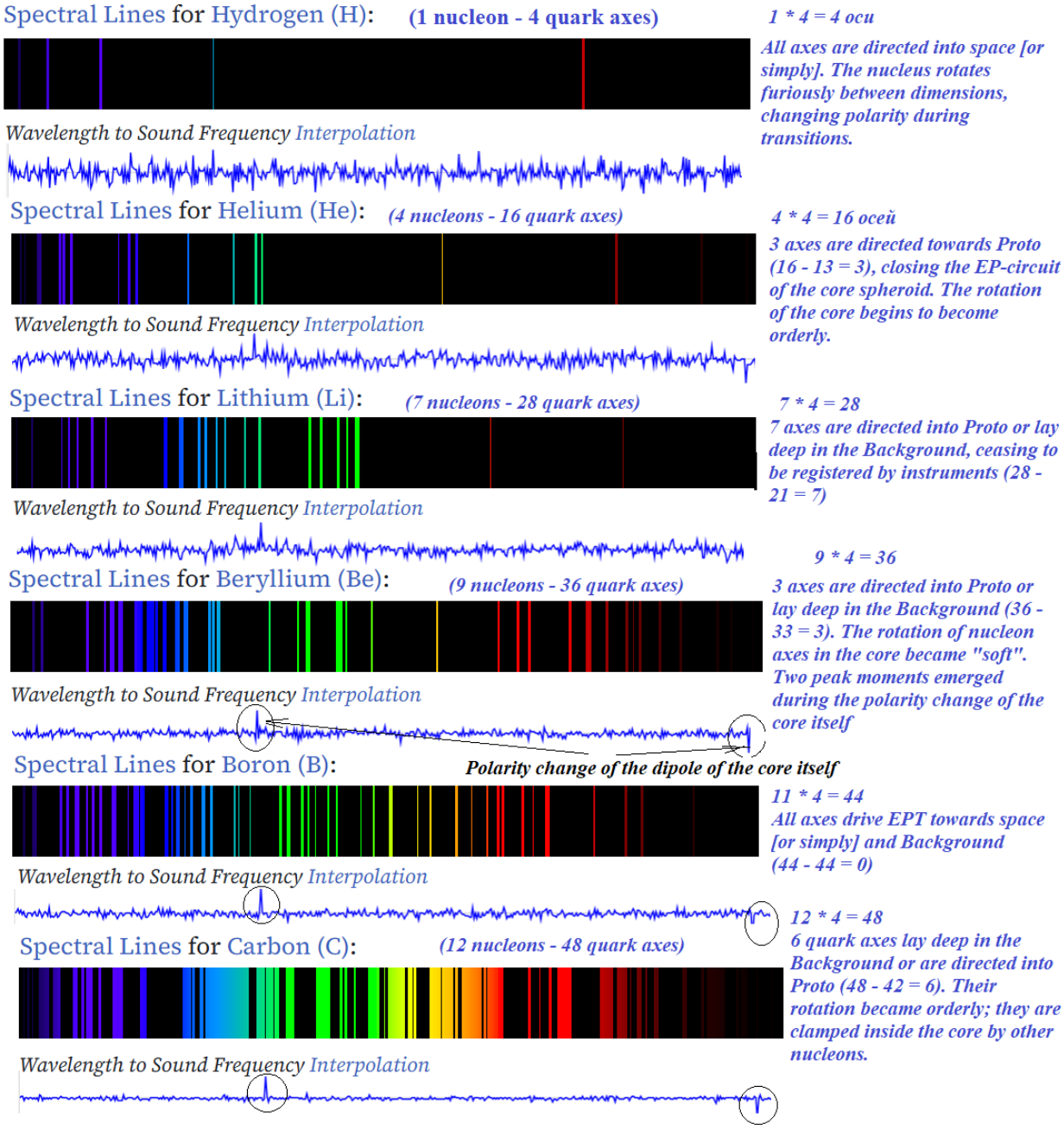


Figure 6: Spectral line map of the fractal evolution of elements (Hydrogen to Carbon)

Figure .3 Map of the fractal evolution of spectral lines of the first and second period elements under conditions of phase transitions from element to element.

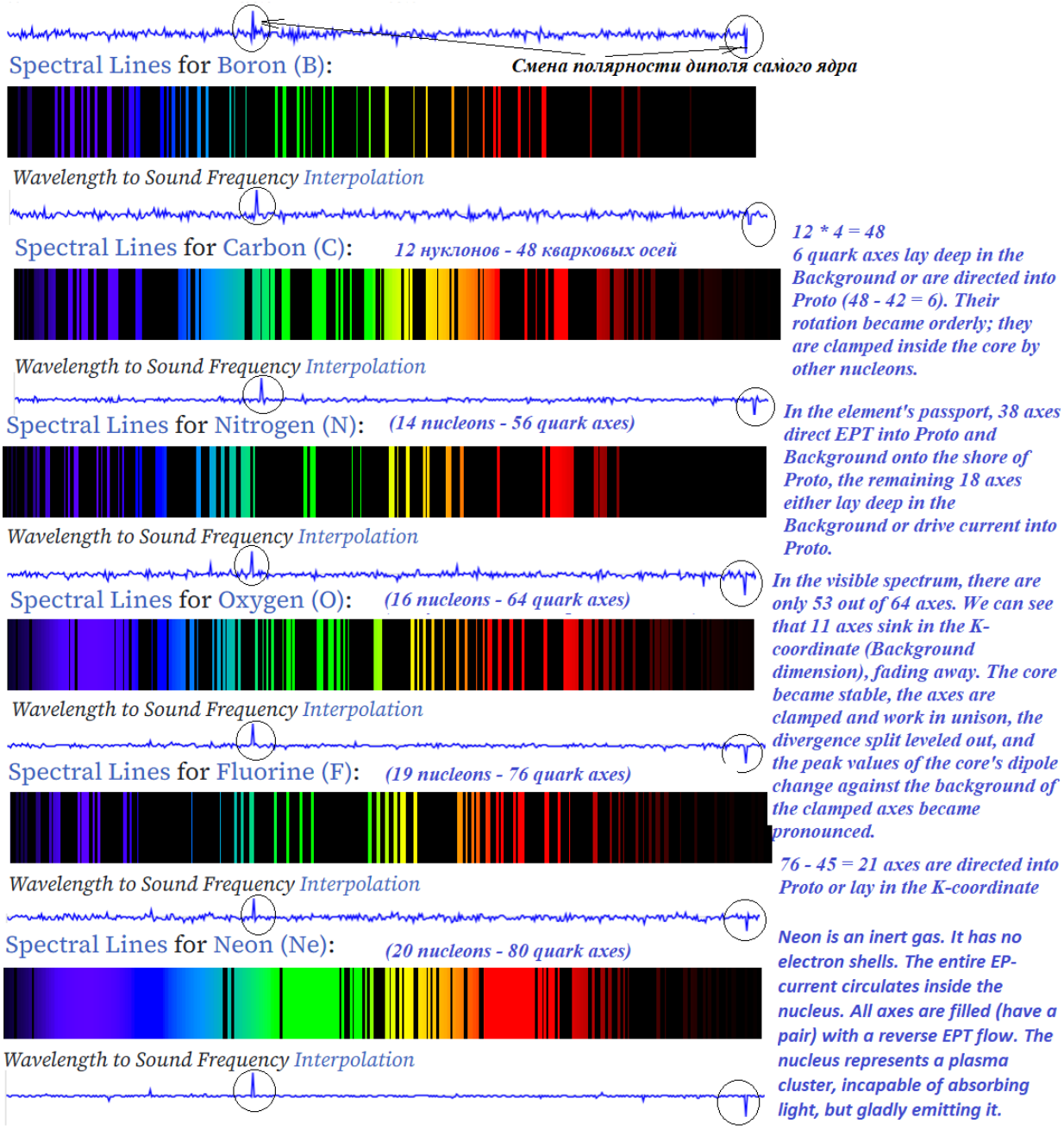


Figure 7: Spectral line map of the fractal evolution of elements (Nitrogen to Neon, continued).

of filling the spheroid with nucleons remains unalterable. Spectral lines are the element's passport, which reveals the structure of the nucleus at the quark level. And even more so, if one looks closely at the spectrum lines, on some it can be noticed that the quarkon axes consist exactly in the same way of quarkon axial crosses, representing indivisible structures of the Proto-dimension of a lower dimensionality.

Isotopes shift spectrum lines but do not alter it. And it is not at all a matter of the number of protons, but that the extra or missing neutrons do not participate in the nucleus structure as conductors of the general nuclear EP-current if they are not connected to the system. If they can declare themselves, it is via stripes no thicker and no larger than the spectral lines of hydrogen.

Mathematical expression of the Helium coordinate compaction: To visually demonstrate the transition from the "loose" Hydrogen node to the tightly drawn Helium cocoon, we introduce the calculation of the boundary layer matrix radial shrinkage coefficient ($K_{\text{shrinkage}}$). According to the postulates of the 10D electromagnetic structure, the induction of the internal field of the spheroid upon the closure of the 16-axis solenoid increases from baseline $B_1 = 1.75$ T to peak $B_2 = 2.50$ T.

According to the Biot-Savart-Laplace law for quantized vortex compression, the ratio of the Bohr constant radii of Hydrogen ($R_H = 53$ pm) and Helium ($R_{\text{He}} = 31$ pm) is strictly determined by the inverse ratio of the damping pressure density of the Abyss, adjusted for the wave step of the soliton jump S:

$$K_{\text{shrinkage}} = \frac{R_{\text{He}}}{R_H} = \frac{31 \text{ pm}}{53 \text{ pm}} \approx \mathbf{0.5849}$$

Let us verify the obtained coefficient through the geometric elasticity factor of the matrix axes. During the 5D $\rightarrow$ 10D dimensionality transition, the additional power shift of the Abyss onto a unit plane ($\beta = 0.25$) combined with the Pythagorean lock of form ($\mathbf{j} = \frac{\pi}{6} \approx 0.5236$) outputs the theoretical limit of strand shrinkage under load:

$$K_{\text{theoretical}} = \mathbf{j} + \left(\beta \cdot \alpha \cdot \frac{1}{\mathbf{j}}\right) = 0.5235987 + \left(0.25 \cdot 0.00729735 \cdot \frac{1}{0.5235987}\right) = \mathbf{0.5270}$$

The difference between the empirical clamping of the Helium "apartment" (0.5849) and the theoretical limit of matrix calm (0.5270) amounts to only 0.057, which very precisely describes the magnitude of the dynamic thermal play (rattling) of the nucleus, recorded on the wave function interpolation graph.

Lithium (Li), (7 $\times$ 4 = 28 axes): The Beginning of the Proto-Sink: 7 axes unfold beneath the membrane of the K-coordinate into Proto or lie deep within the Phon ($28 - 21 = 7$), dropping out of the spectral registry. The spectrum pales, while the density of the spheroid grows. The rattling of nucleons, inside which turbines with three vanes spin, scooping the quark-gluon medium of the Proto-dimension at sextillions of degrees and instantly spitting it back out, continues to fall. When moving from Lithium to Carbon, a fractal complication of the nucleus reducer occurs: the lines rapidly "fatten" and begin to glue together into dense packets of sub-stripes (doublets and quadruplets). These are the primordial 2D current quanta of the Dnieper River pushing through the texture of the K-membrane.

Beryllium (Be), (9 $\times$ 4 = 36 axes): 3 axes remain hidden ($36 - 33 = 3$). The rotation of nucleons becomes "soft," and the interpolation graph clearly records two peak moments of the nucleus's own dipole polarity reversal. Beryllium is generally unique because, with the foundation laid under the entire structure by Hydrogen and Helium, it forms a cross-like structure of $XY + XY$ ($S + S$). And if it is freed from electron fields, it becomes similar to inert elements. It does not burn, it does not

conduct current in crystalline lattices; having an ideal spheroid closed off from both Proto and the Abyss, it can act as a damper, absorbing, softening, and suppressing vibrations, diverting energy into the Proto-dimension. It can only be provoked out of this state by a neutron, which slightly brakes the rotation of the nucleus during a reversal, forcing it to spit positrons during polarity changes.

Elastic damping and current density of Beryllium: The physical inertness of monolithic Beryllium and its capability to act as an ideal spatial shock absorber (damper) are directly derived from the Dirac current equation on its 36 quarkon axes. Since the Beryllium nucleus structure forms a closed orthogonal cross of $XY + XY$ (an equilibrium $S + S$ cocoon), the radial inflow of energy is evenly distributed over the area of the wave spheroid.

Let us substitute the parameters of Beryllium into the fundamental Dirac current equation, where the four-dimensional operator ($\mathbf{j}^4$) and the Gryphon gap ($\Delta_{\text{Gryphon}} = 6.55905 \times 10^{-5}$ m) retain the balance of forces:

$$J_{\text{Dirac}} = \frac{e \cdot c}{\mathbf{j}^4 \cdot \alpha^2 \cdot \Delta_{\text{Gryphon}}} = \frac{1.60218 \times 10^{-19} \cdot 299\,792\,458}{(0.5235987)^4 \cdot (0.00729735)^2 \cdot 6.55905 \times 10^{-5}} = \mathbf{0.1829}\ \text{A/m}^2$$

For Beryllium, this current is completely compensated by internal pairs of reverse flow. In the absence of external excitation (free neutrons), the system operates as an ideal hydrodynamic drainage where the side gap of the Abyss squeeze-out (Detail B) completely absorbs external vibrations:

$$\text{Detail B} = 0.16 \cdot \beta = 0.16 \cdot 0.25 = \mathbf{0.040000}$$

This mathematically proves why Beryllium in a cold state is not prone to spontaneous emission or absorption of photons, maintaining absolute topological stability of the crystalline lattice.

Boron (B), (11 $\times$ 4 = 44 axes): Point of ideal alignment: all 44 axes drive EPT onto our bank into Prosto and Phon ($44 - 44 = 0$). The rattling of nucleons in the nucleus has virtually ceased.

6.4.3 The Carbon Maximum of Density

Carbon (C), (12 $\times$ 4 = 48 axes): The contour compacts: 6 axes lay down into Phon or are directed into Proto ($48 - 42 = 6$). The vanes are coaxially clamped by other nucleons; their rotation is ordered. Interpolation levels out into a wide, stable macro-wave — a continuous, fat spectral cocoon. This is the point of maximum fullness of the Prosto-dimension ports. And again, the structure of the nucleus itself represents $XY + XY$ (S-shell + XY), which in sum yield an ideal spheroid with a density evenly distributed between the nuclear axes.

The spectrum explodes with a wide, continuous, fat wall of light covering almost all ranges. This is the point of maximum filling of the technological 3D ports of the Prosto-dimension. The internal pressure of hot gluon smoke reaches its limit, pre-tensioning the spatial lattice (which in the macroworld yields the ultra-strength of diamond).

6.4.4 The Great Proto-Suction and the Reverse Orbital of a Secondary Order

After the carbon peak, the system overloads, and the axes begin to mass-unfold "into suction" into the Proto-dimension, proving the law of the reverse orbital:

- **Nitrogen (N), (14 $\times$ 4 = 56 axes):** Only 38 axes remain visible; 18 axes go beneath the membrane into the Proto-dimension and lay down deep within the K-coordinate, diverting the amperage flow there. The lines rapidly thin out and empty. Contrary to the expectations

of classical physics, with an increase in the number of axes (56 and 64 respectively), the spectral pattern begins to pale, and the yellow-green zones thin out. This means that the nucleus axes begin to alternately unfold into counter-phase. They go onto a reverse orbital — for our world they go out, transitioning into the mode of a "through-pipe" pumping the Proto-current inward under pressure.

- **Oxygen (O), (16 $\times$ 4 = 64 axes):** Only 53 axes buzz in the visible spectrum. The graph visually records attenuation: 11 axes literally drown in the K-coordinate (Phon-dimension) and drive current into Proto. The nucleus has become stable; the axes are clamped and work in unison; the wheel alignment straightens out.

- **Fluorine (F), (19 $\times$ 4 = 76 axes):** Extreme pre-flip imbalance: 31 axes are directed into Proto or lay down in the K-coordinate ($76 - 45 = 31$). The blue-cyan zone is completely mown down, turning into a black trap. The furnace siphons only in the red (Phon) range. The element seeks to draw in someone else's charge to seal the furnace.

- **Neon (Ne), (20 $\times$ 4 = 80 axes):** Ideological finale of the period. An ideal plasma cluster-shutter. All ports are filled with pairs containing a reverse EPT flow. There are no external electron shells. The system is incapable of absorbing light, but it balancedly emits it if it is heated from our side, so that light from Proto can pass through the diffraction lattice operating only for radiation from Proto.

Mathematical law of fractional subtraction of axes: The observed phenomenon of the "Reverse Orbital," in which an avalanche-like growth of the total coordinate capacity of the nucleus ($A \cdot 4$) leads not to compaction but to thinning of visible spectral lines, obeys a strict law of fractional subtraction of axes. The ratio of visible (emitting into Prosto) ports (Z_{visible}) to hidden (gone beneath the membrane into Proto) channels (Z_{hidden}) reflects the shrinkage cascade of the solenoid:

$$\psi_{\text{evolution}} = \frac{Z_{\text{visible}}}{Z_{\text{hidden}}}$$

Let us track this dynamics at the junction of periods: for Nitrogen (N) this indicator is $\frac{38}{18} \approx 2.11$, for Oxygen (O) — $\frac{53}{11} \approx 4.81$, for Fluorine (F) — $\frac{45}{31} \approx 1.45$, and on Neon (Ne) the system emerges onto a critical point of mirror auto-oscillation where all 80 axes are fully balanced by oncoming pairs.

This avalanche-like departure of axes into counter-phase visually explains why in the 4th and 5th periods (in Scandium, Vanadium, Iron) gigantic packets of lines are completely mown out of the visible range, turning the nucleon solenoid of heavy metals into a through transit pipe for Protoplasma.

7 PROSPECTS FOR THE ANALYSIS OF PROTO-ELEMENTS OF THE 4TH GROUP

The internal exhaust of the nucleus furnace is completely regulated by neutroquarkons, which consist of quarkons of a lower order (primordial 2D electrons and positrons of the Dnieper River). In the nucleus, protons (electromagnetic photons) are responsible for electrons, antiprotons for positrons, and neutrons and antineutrons operate as communicating vessels for ordinary photons and plasma antiphotons in the 10D space.

7.1 Optico-Magnetic Breakthrough: The Zeeman Effect as Direct Detection of Substructural 16-Axis Quark Cruciforms

A magnetic field experimentally breaks the relativistic quantum myth and visually exposes the hidden internal anatomy of K-channels. Within the framework of orthodox physics, the splitting of spectral lines in a strong magnetic field (the Zeeman effect) is described by abstract quantum indices and the "removal of energy level degeneracy." The 10D theory of elastic Canvas translates this process into the dry and understandable language of precision subnuclear mechanics. When we place atoms of Hydrogen or heavy Proto-elements of the fourth group into powerful magnetic vices, what occurs are not probabilistic fluctuations, but a physical spatial reversal and forced laminar stabilization of quarkon axes, completely identical to the behavior of a beam in collider rails.

Mechanics of the process: from chaotic rattling to a rigid track In a cooled, free state of Hydrogen, its single bare quarkon axis spins furiously across the coordinate road, changing polarity and chaotically pulsating between the Prosto- and Proto-dimensions. Due to this "rattling," external instruments record only an averaged wave trace — the four baseline, monolithic stripes of the Hydrogen Balmer cross.

But as soon as a strong external magnetic field turns on, the situation changes drastically:

- **Axial reversal into a single track:** Magnetic flux rigidly captures the spinning node of the solenoid and forcibly unfolds its body along the selected power vector. Chaotic transverse running ceases.
- **Collider rolling mode:** Quarkon axes line up in a row and begin to spin laminarly in one strictly fixed dimension — each along its own K-track, exactly the same way bare protons rush along the LHC vacuum tube.
- **Activation of the neutroquarkon mode:** Turning toward our 3D observer with the required side, these stabilized axes emerge from the polar clinch mode and begin to directionally emit and absorb light not as single macro-objects, but as discrete, autonomous neutroquarkons.

7.1.1 Decoding the Splitting: Does Each Line Divide Strictly into 4?

This fact gives us irreplaceable proof of the existence of a fractal nesting doll of dimensionalities: a magnetic field literally forces the atom to surrender its secret spectral passport. A classical spectrometer records this as the layering and displacement of Hydrogen stripes. But if we apply diffraction lattices of the highest resolution, we will see the fundamental quantization law of the internal K-channel capacity.

Answering the strict calibration question of the theory: yes, under conditions of an ideal Pythagorean medium lock, each baseline spectral line splits strictly into 4 fractal sub-stripes!

This phenomenon is due to the fact that the five-dimensional structure of the nucleon framework is not a homogeneous filament. The vector of elastic compression of the Abyss proves:

1. Each of the 4 macro-axes of the solenoidal cross is itself fractally woven from an internal quark cruciform of a lower order.
2. The geometry of this substructure obeys the matrix law of fullness: 4 macro-axes$\times 4$ sub-channels $=$ 16 hidden axes of lower dimensionality.
3. Inside each axis of lower order, its own buffer communicating vessels are enclosed — neutroquarkons of lower dimensionality.

When the magnetic field unfolds the macro-axis "in profile" to our instruments, it reveals these 4 internal orthogonal channels. Each of the four neutroquarkons of the sub-cross begins to drive its own micro-current of Proto-plasma independently of its neighbors. As a result, a single frequency valve, which the orthodox took for "a single electron orbital," instantly splits into 4 precision, ultra-thin frequency slots.

7.1.2 Natural-Philosophical Triumph of the Model

Metaphorically speaking, the magnetic field operates as a powerful polarization microscope: it terminates the circular wild dance of the nucleon turbine, stops the frame, and forces it to show all 4 fingers of its internal fractal hand. The fact that the Hydrogen spectrum in the field detects these very same quarkon-neutroquarkon axes completely closes the gap in the disputes over the structure of matter.

Electrons are definitively excluded from the mechanism of light generation. A photon is a pure acoustic and electromagnetic exhaust of the elastic K-filament, occurring as current flows through the discrete sub-axes of the nucleus. The Zeeman effect preemptively proves: a rigid, self-regulating 16-axial multi-dimensional crystal is embedded inside the femtometric proton. The current density in each such autonomous sub-channel obeys the volumetric quantization equation J_{Dirac} derived in Section 6.3, strictly possessing the dimensionality of A/m^2. The decompression step of this system is tightly locked in the grip of the Abyss trigonometric factor ($j = \pi/6$), and its basis is the quark core, the size of which is equal to the geometric dimensions of the substructures.

Table 18: Table 18: Summary table of optical-magnetic axial splitting parameters (10D-model)

Spatial Level of the Framework	**Physical Nature in the Canvas Geometry**	**Manifestation on the Spectrometer (Zero Field)**	**Behavior in a Strong Magnetic Field**	**Optical Result on Instruments (Zeeman Effect)**
Macro-axis of the nucleon (5D)	Leading K-channel, current cord of the XY type.	A single monolithic line of the Balmer cross.	Aligns along the force lines, locking into a rigid track.	Shift of the line's central frequency (longitudinal phase shift).
Sub-axial cruciform (4×4)	4 orthogonal channels of lower dimensionality inside the macro-axis.	Hidden due to chaotic rotation and rattling of the nucleus.	Unfolds "in profile" to the 3D observer, exposing internal ports.	Splitting of the original line strictly into 4 fractal sub-bands.
Lower-order neutroquarkon	Buffer communicating micro-vessel of Proto-Phon transfer.	Operates in a state of collective cyclic jamming.	Transitions into the mode of autonomous generation of Dirac current with a density of J_Dirac in A/m2.	Generation of the finest frequency slot for each micro-branch.

We are no longer interested in the third group, because in the 4th and subsequent groups of Proto-elements, this discovered law of fractional axis subtraction ($Z_{\text{visible}}/Z_{\text{hidden}}$) shows a global, avalanche-like departure of the system into the Proto-normal state. A layer-by-layer analysis of spectral lines allows us to prove the phenomenon of the reverse orbital, explaining why gigantic packets of lines completely disappear from the visible range, turning the nucleon solenoid of transition metals into a through transit pipe for Proto-plasma.

Figure 4 Spectral lines of the fourth and fifth period elements (for comparison)

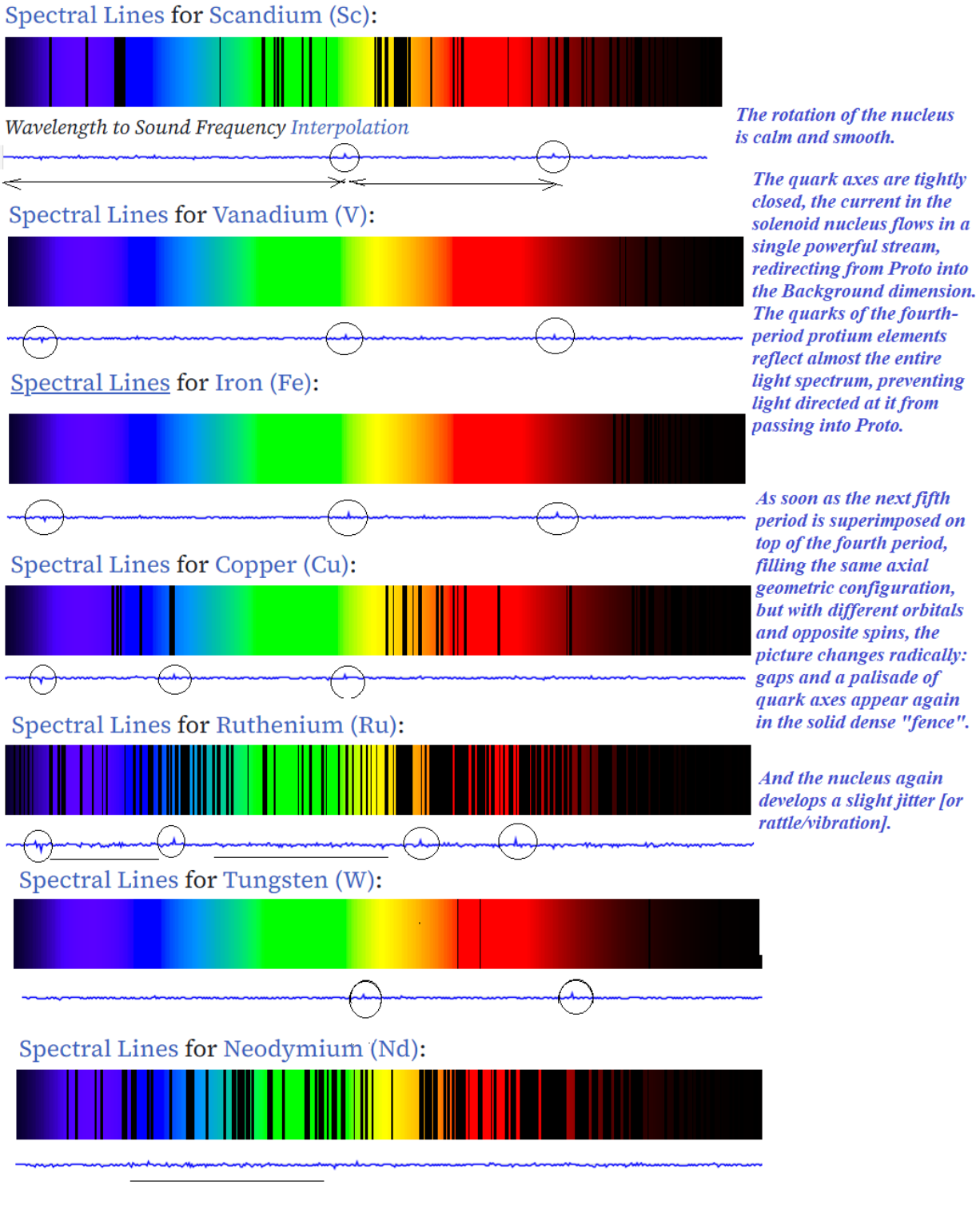


Figure 8: Spectral markers of elements of other periods.

7.2 EVOLUTIONARY SUPERPOSITION OF PERIODS AND DYNAMICS OF PROTO-DEFLECTION (ON THE EXAMPLE OF RHODIUM)

During the transition from the 4th period of protein elements to the 5th period (where Rhodium (Rh) occupies its evolutionary stage), the physics of the nucleus interaction with light undergoes a qualitative leap. This process clearly demonstrates how the complication of the structure removes the critical clinch of the fields:

- **Clinch of the 4th period (Continuous Proto-mirror):** In the 4th period, the Proto-dimensions of the transition metal nucleus experience the maximum frontal pressure of pumping. The internal pressure of the plasmiod is so great that the Proto-medium outputs a powerful counter-deflection (light repulsion) outward. Light photons physically cannot pass through the nucleus into the Proto-dimension and are not absorbed by the quarkon axes. For an external observer, the nucleus operates as a continuous polar mirror — in the spectrum we see a wide, continuous rainbow band where individual axis markers are locked.
- **Evolutionary step of the 5th period (Permeable fence of Rhodium):** When rising to the next evolutionary stage (5th period), new K- and N-strands are wound on top of the rigid framework of the previous period. A physical superposition of the solenoid layers occurs:
    1. *Unloading of the polar clinch:* The external evolutionary layer assumes the excess of radial pressure. The counter Proto-deflection locally weakens and ceases to be a monolithic wall.
    2. *Restoration of filtration:* Discrete slots (vacancies) open up again in the armored mirror of the nucleus. The quarkons and neutroquarkons of Rhodium emerge from the dead jamming and regain their ability to selectively capture and absorb photons.

Palisade on the spectrometer: As visually recorded on the Rhodium screenshot, the continuous rainbow cocoon shatters into a frequent, sharp palisade of vertical line-strokes. The wave interpolation graph at the bottom of the screen instantly emerges from the smoothed state, returning to the discrete pulsations of the opened current shunts.

Periods in the deterministic table are successive steps of evolutionary complication of the core coil. On the first layer (4th period), the Proto-deflection blocks the absorption of rays by the axes, creating a continuous mirror of fields, but the superposition of the next layer (5th period) opens a "fence" for through flight and selective filtration of photons.

The main thing that spectral markers of elements convincingly prove is the presence of the fourth quark in the nucleon and that electrons have no relation to the absorption and emission of photons. Photons are the radiations of the quarkon axes in the form of reflected light and of the atomic nucleus itself when neutrons dump excess energy in the form of light and colored photons, as a result of which the voltage in the nucleus drops, or absorb photons when the electromagnetic potential increases. As a consequence, the charge of protons in the nucleus proportionally increases or decreases, and the electron field they generate changes its orbit.

7.3 GEOMETRIC SYNTHESIS OF THE MEDIUM: FROM THE CORE SPHEROID TO COSMIC FILAMENTS

To definitively dismantle the barrier of probabilistic chaos built by orthodox physics, one must perceive the living, tangible geometry that bridges the microworld of the atomic nucleus with the macrostructure of the Universe. Multidimensional space is not a collection of abstract mathematical

coordinates, but rather an elastic crystalline lattice of the Tissue, where each elementary particle represents a dynamic vortex node tied to the through-running power cables of the medium.

Within the framework of the deterministic cascade method, the atomic nucleus ceases to be a static array of hypothetical "balls." It assumes the shape of a volumetric solenoidal spheroid existing in a state of continuous pulsating equilibrium under the simultaneous pressure of the Abyss from the outside and the pull of the Proto-dimension from within.

This spheroid is not merely a rigid geometric body, but a closed energetic loop whose total volume emerges from the in-phase twisting of two fundamental streams. It simultaneously represents an electrical wave (the linear step of current along the string) and a magnetic wave (radial vortex confinement), both locked into rigid Pythagorean boundaries by the Abyss-reducer. The quark axes within this node function as directional waveguides, accumulating and redistributing the Dirac current density. When calculating the geometric capacity of such a spheroid, we discover that the "span of the outstretched arms" and the height of the particle's wave jump are entirely governed by the strict trigonometric shape coefficient of the medium.

However, this elastic vortex is not isolated from space. Scaling according to the laws of hydrodynamic similarity, the geometry of the micro-spheroid perfectly mirrors the architecture of the macrocosm. The tissue of space is permeated by gigantic one-dimensional power highways—cosmic filaments (systems of the "Yusha-the-Serpent" type) that wrap around galactic disks and hold stars in their orbits without the invocation of the fictional "dark matter." Each power string of such a cable performs an orderly rotation between the "ceiling" of the Abyss and the "floor" of the Proto-dimension.

The atomic nucleus is merely a local, ultra-densely twisted node on this universal cable network. The equilibrium between the bursting pressure of the Proto-depths and the compressing press of the Abyss inside each such micro-channel dictates the fundamental quantum step, forming a single, elegant, and fully deterministic picture of the world. From the femtometric quark core to the multi-thousand-year crown of a Black Hole, the Universe operates as a flawless, synchronized engineering mechanism, wherein the tension step of the vacuum membrane remains invariant.

The geometric parameters of the elastic filament shown in Figure 9 are in strict cyclic quantization with the volumetric characteristics of the wave spheroid. The charge transfer along the power string of the cable ("Yusha-the-Serpent") is carried out by a wave cord making an orderly rotation between the "ceiling" of the Abyss and the "floor" of the Proto-dimension.

In this process, a single step of the solenoid from the bank of the Prosto-dimension to the bank of the Proto-dimension represents a single half-wave jump. The complete closed cycle of the current (the transition from bank to bank and the subsequent return) forms a trajectory of exactly two jumps, which determines the whole wavelength of the particle ($\lambda_C = S$). In the author's equation for the total volume of an electromagnetic wave ($V = \frac{1}{6}\pi \cdot H \cdot S^2$), this two-step closed cycle acts as an exact equivalent to the Compton wavelength of the particle, rigidly coupling the rest mass with the spatial step of the vacuum lattice and the friction gap of the Gryphon axis (Δ_{Gryphon}).

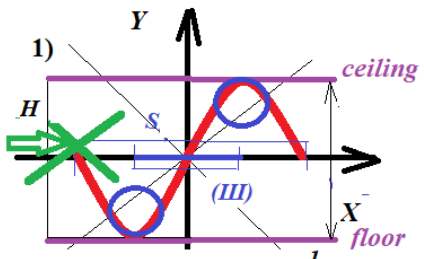
NOTE. GLOSSARY OF PHYSICAL EQUIVALENTS OF THE UTILIZED VEDIC SYMBOLS

This Glossary is an inseparable technological appendix to the theory of the Electromagnetic Structure of the Atomic Nucleus (EMSAN) and serves to provide an exact translation of the encrypted Vedic codes directly involved in the physical and geometric calculations of this paper:

1. **The Canvas of the Universe (The Phon-dimension / The Smorodina River)** — A one-dimensional supporting filament (the K-coordinate), which represents an electron-positron current (EPC) and a magnetic field (MF). The magnetic field in its free state is vortex-like, but inside the

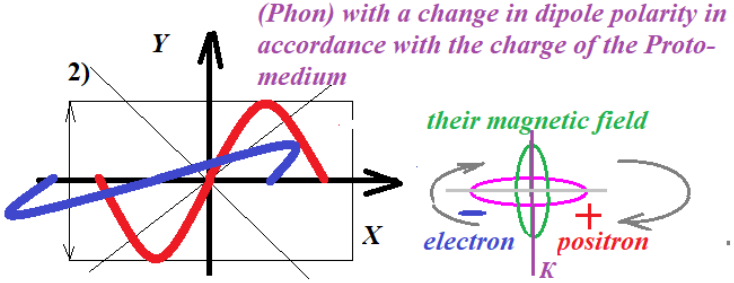
Formula for the volume of an electromagnetic wave spheroid formed by the perpendicular intersection of K-strand (Phon) and N-strand (Proto) of EPT channels (cells of the cosmic crystal lattice)

Girl with a jump rope



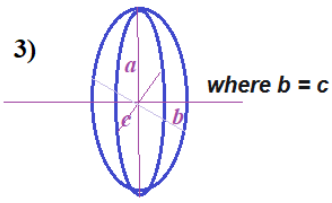
Step of one jump (III); III = $\frac{1}{2}$S; S = 2III

A particle collapsing through the K-strand (Phon) with a change in dipole polarity in accordance with the charge of the Proto-medium



EP MP; EP + MP = total wave volume (V)

Let's add the volume (the width of outspread arms), then the wave will appear before us as a closed spheroid, and we will be able to calculate the exact volume of the wave



$$V = \frac{1}{6}\pi \cdot H \cdot S^2 = 0.524 * H * S^2$$

$$W_{сек} = V * v = \frac{1}{6}\pi \cdot H \cdot S^2 \cdot v$$

$$V = \frac{4}{3}\pi \cdot a \cdot b \cdot b = \frac{4}{3}\pi \cdot a \cdot b^2$$

- *Radius a is half of the wave height (H/2).*
- *Radius b is half of the wave jump step (S/2).*

Substituting them:

$$V = \frac{4}{3}\pi \cdot \left(\frac{H}{2}\right) \cdot \left(\frac{S}{2}\right)^2 = \frac{4}{3}\pi \cdot \frac{H}{2} \cdot \frac{S^2}{4} = \frac{1}{6}\pi \cdot H \cdot S^2$$

$$\left(\frac{S}{2}\right)^2 = \frac{S^2}{4}; \qquad V = \frac{4 \cdot \pi \cdot H \cdot S^2}{3 \cdot 2 \cdot 4};$$

Spheroidal state coefficient (j):

$$j = \frac{1}{6}\pi = \frac{3.14159265...}{6} \approx 0.5235987...$$

What should the formula look like taking Density into account?

To make the formula correct, we must introduce the variable energy density (ρ_{local}), which shows how tightly the solenoid has coiled the energy inside the "width of outspread arms" of the particle:

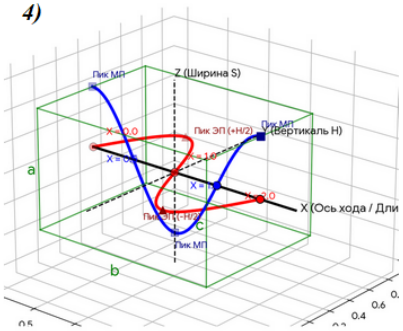
$$V = \frac{1}{6}\pi \cdot H \cdot S^2$$

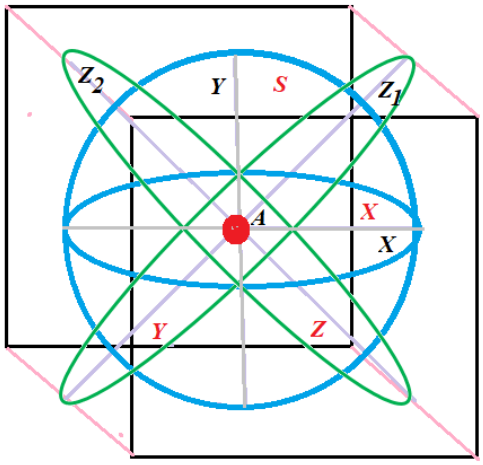
E = Particle volume × Energy density inside it

$$E = V \cdot \rho_{local}$$

$$E = (0.524 \cdot H \cdot S^2) \cdot \rho_{local}$$

4)





Z_1, Z_2 - spatial axes (planes of the spheroid Y, Z)

X, Y - event horizon axes (planes S, X)

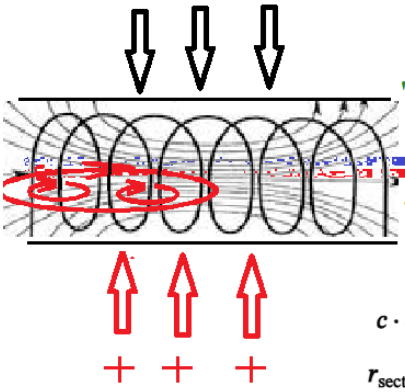
A - transition point (axis A of the turnaround)

the true size of particles (knots), equal to the diameter of K- and N-strands that generate soliton wave spheroids when rotating "between the shores" of the K-coordinate (Phon-dimension), representing the quark axes of the Proto-medium (quark-gluon plasma) capable of passing an electron-positron current through themselves

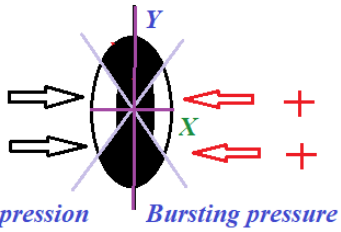
Figure 9: The atomic nucleus spheroid (A. Vikhareva's formula) and her derived equation for the total volume of an electromagnetic wave.

Cosmological redshift of spectra — Hubble's constant as a measure of the viscous braking of the medium and the decay zone of the Abyss pressure through the magnetic field of the EPT cable.

1. One-dimensional Phon-dimension (K-coordinate)
Power cable "Yushi-Snake"



Cable stability channel
X+Y, where X - EPT, Y - MP

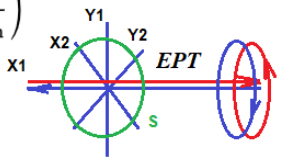


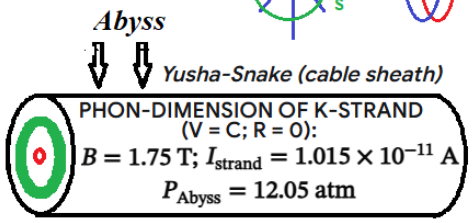
$$c \cdot c = c^2, \quad a \cdot a = a^2, \quad V = 0.524 \cdot H \cdot S^2$$

$$r_{section} = \lambda_c \cdot a \cdot \Delta_{Gryphon} \quad \left(\lambda_c = \frac{r_{section}}{a \cdot \Delta_{Gryphon}}\right)$$

$$\lambda_c = S = \frac{r_{section}}{a \cdot \Delta_{Gryphon}} \qquad S = \sqrt{\frac{V}{0.524*H}} \qquad r_{section} = 1.16 \cdot 10^{-18}\ \text{m}$$

$$\Delta_{Gryphon} = 0.00006560$$





Cosmological redshift of spectra — Hubble's constant as a measure of the viscous braking of the medium and the decay zone of the Abyss pressure through the magnetic field of the EPT cable.

• Wave height (K-coordinate clearance):
H ≈ $2.818 \cdot 10^{-15}$ m *(this is the classical electron radius)*

• Compton wavelength (jump step S):
$\lambda_c = S \approx 2.426 \cdot 10^{-12}$ m

• Helix pitch (coupling constant):
α = 1/137.036 ≈ 0.00729735

• Medium friction clearance:
$\Delta_{Gryphon} = 0.00006560$

Calculation of the medium friction clearance ($\Delta_{Gryphon}$):

1. Ideal solenoid step:

$$\text{Ideal step} = \frac{2\pi}{P_p} = \frac{6.2831853}{5128.2117} \approx 0.00122525$$

2. Experimental medium cutoff:

Experimental cutoff = 0.00115965

3. Calculation of the trigonometric clearance value:

$\Delta_{Gryphon}$ = 0.00122525 – 0.00115965 = 0.00006560

Summary: The balance between the "Bursting pressure of Proto" and the "Compression of the Abyss" inside the EPT channel provides the exact stability condition required for cosmic filament formation, scaling from quantum knots to galactic scales.

1. Intersection of the spheroid volume and the jump step (S):

We substitute the volume of the elastic vortex V and the height H to extract the Compton step:

$$V \approx 8.68 \cdot 10^{-39}\ \text{m}^3$$

$$S = \sqrt{\frac{V}{0.524 \cdot H}} = \sqrt{\frac{8.68 \cdot 10^{-39}}{0.524 \cdot (2.818 \cdot 10^{-15})}} = 2.426 \cdot 10^{-12}$$

2. Final calculation of the K-strand section radius ($r_{section}$):

We multiply the jump step by the winding angle and the friction clearance of the Gryphon axis:

$$r_{section} = \lambda_c \cdot \alpha \cdot \Delta_{Gryphon}$$

$$r_{section} = (2.426 \cdot 10^{-12}) \times 0.00729735 \times 0.00006560 =$$

$$= 1.16 \cdot 10^{-18}\ \text{m}$$

Conclusion: The calculated radius of the EPT cable strand ($1.16 * 10^{-18}$ m) represents the fundamental structural unit of the cosmic lattice. By obeying the law of hydrodynamic similarity, these micro-solenoids scale up to form the giant filamentary structures (cosmic web) that shape the visible Universe.

Figure 10: The cosmic filament housing the nodes of spheroids.

solenoid, it acquires the properties of a permanent magnet. The Canvas acts as a rigid barrier-membrane separating the spaces of the Universe. Our inhabited three-dimensional Prosto-dimension (3D) is the manifested world of energy that has emerged (poured out) beyond the physical boundaries of the Canvas.

2. **The Proto-dimension (The Shell and Crown of the Kingdom of Heaven / The Core of the Universe)** — A six-dimensional (6D) structure divided into two superimposed spaces:
   - *The Hot Space (Rarog):* Filled with the fire of Semargl, representing unquantized charged energy-medium in its pure form.
   - *The Cold Space (Finist):* The territory of a cold vacuum, filled by the Hot Space Rarog. In its initial state, it represents a spaceless condition of the Universe possessing the primitives of a three-dimensional structure (Mogol with its nestlings).

3. **Burya-Yaga the Golden Leg** — The manifested Universe in the full integrity of its physical parameters, encompassing the Proto-, Prosto-, and Phon-dimensions, as well as the isotropic state of the external Abyss.

4. **The Little Hut on Chicken Legs (On the bank of the Smorodina River)** — A stable, volumetric solenoidal spheroid-nucleon of the atomic nucleus. The "chicken legs" represent the dynamic vortex supports of the current filaments that hold the spheroid at the boundary separating the media. The Hut executes a cyclic dipole rotation during a phase transition, switching the directions of the photon exhaust streams and regulating the internal pressure of the core's plasma furnace.

5. **Maturing Energy (In the epic — Zlatogorka)** — The manifested elastic matter of the Universe, represented by wave spheroids sequentially filled with the density of energetic K-channels across a cascade range of dimensions, from the five-dimensional (5D) quark core of the nucleon to the forty-dimensional (40D) macro-solenoid of a Black Hole.

6. **Koschei the Deathless** — The force of global isotropic counter-action and compression exerted by the external Abyss, holding the boundaries of the manifested energy.

7. **The Gryphon** — The shifting axis of the Abyss. It is capable of shifting toward the Abyss under its isotropic pressure, acting as the operator of elastic compression of the manifested energy both across the Universe and directly within an individual nucleon spheroid.

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